.

The saturation theorem is a chart-coordinate statement on $U_1$: the scalar $\kappa(\mathcal{A}) = \mathrm{av}(r_{\mathcal{A}}(z))$ is the unique $\Sigma_n$-coinvariant datum that survives averaging from the ordered to the symmetric bar, and it determines the degree-two universal Maurer–Cartan element at the $\Phi_{\mathrm{KZ}}$ base point.

# KOSZULNESS MODULI WITH FOURTEEN ATLAS CHARTS: THE KONTSEVICH–TAMARKIN REFORMULATION OF CHIRAL KOSZUL DUALITY

RAEEZ LORGAT

ABSTRACT. For a chiral algebra $\mathcal{A}$ on an algebraic curve $X$, equipped with its PBW filtration, we construct a moduli scheme $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$ whose $\mathbb{C}$-points are the Koszulness characterizations $(\Phi, c_\Phi)$: a pair consisting of a Drinfeld associator $\Phi$ and a chart coordinate $c_\Phi$ such that $\mathcal{A}$ is chirally Koszul if and only if the chart predicate $\Pi_\Phi(\mathcal{A})$ holds.

The scheme has four structural properties. (A1) $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A}) \neq \emptyset$ if and only if $\mathcal{A}$ is chirally Koszul. (A2) $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$ is a torsor for the Grothendieck–Teichmüller group $\mathrm{GRT}_1$; transitions between charts are Tamarkin $\Phi$-transfer maps. (A3) The fourteen classical characterizations form a finite affine atlas $\{U_j\}_{j=1}^{14}$: ten charts $U_1, \ldots, U_{10}$ at the Knizhnik–Zamolodchikov associator $\Phi_{\mathrm{KZ}}$ (the core ten of [23]), one chart $U_{11}$ at the Alekseev–Torossian associator $\Phi_{\mathrm{AT}}$ (Lagrangian chart; the perfectness hypothesis is killed by the AT gauge), one chart $U_{12}$ at the de Rham–Betti associator $\Phi_{\mathrm{dR,B}}$ (Hodge–Betti chart; mixed Hodge module purity for Virasoro is closed unconditionally by Feigin–Fuchs BGG resolution), one chart $U_{13}$ at the Enriquez elliptic associator $\Phi_{\mathrm{ell}}$ (genus-one twisted Künneth), and one chart $U_{14}$ at the Kontsevich integral associator $\Phi_{\mathrm{Kon}}$ ($\mathrm{SC}^{\mathrm{ch,top}}$ homotopy-Koszulity as chart tautology). (A4) The shadow-class stratification G/L/C/M/FF selects a distinguished chart on each stratum: $\Phi_{\mathrm{KZ}}$ for class G and the generic Kac–Moody locus, $\Phi_{\mathrm{ell}}$ for class L, $\Phi_{\mathrm{AT}}$ for class C, $\Phi_{\mathrm{Kon}}$ (denoted $\Phi_\infty$) for class M, and $\Phi_{\mathrm{dR,B}}$ for the Feigin–Frenkel locus at critical level.

The reformulation makes explicit the Kontsevich–Tamarkin principle: chiral Koszulness is a $\mathrm{GRT}_1$-invariant *property*; the fourteen classical characterizations are $\mathrm{GRT}_1$-coordinate *charts*. Two consequences close frontier gaps without bar–cobar detour. First, for Virasoro at generic $c$, $H^\bullet(\mathrm{Vir}_c, \mathrm{Vir}_c)_{\mathrm{weight}\, r+1} = 0$ for all $r \geq 2$ by Feigin–Fuchs BGG resolution together with Kac determinant $\det G_h \neq 0$ outside the minimal-model lattice: the $U_{12}$ chart closes the circularity of . Second, for the Yangian $Y_\hbar(\mathfrak{g})$, the associative atlas embeds as an open subscheme, $\mathcal{M}^{\mathrm{assoc}}_{\mathrm{Kosz}}(Y_\hbar) \hookrightarrow \mathcal{M}^{\mathrm{chiral}}_{\mathrm{Kosz}}(Y_\hbar)$; the honest open question is chart-surjectivity (closes )

## CONTENTS



*Date*: April 23, 2026.

**Definition 0.1** (Koszulness moduli scheme)**.** Let $\mathcal{A}$ be a chiral algebra with PBW filtration. The *Koszulness moduli scheme* $M_{\mathrm{Kosz}}(\mathcal{A})$ is the $\mathrm{GRT}_1$-equivariant moduli scheme whose $\mathbb{C}$-points parametrize *Koszulness characterizations* of $\mathcal{A}$: pairs $(\Phi, c_\Phi)$ where $\Phi \in \mathrm{GRT}_1(\mathbb{C})$ is a Drinfeld associator and $c_\Phi$ is a $\Phi$-dependent equivalence statement

$$c_\Phi \;:\; [\mathcal{A} \text{ is chirally Koszul}] \iff [\Pi_\Phi(\mathcal{A}) \text{ holds}],$$

with $\Pi_\Phi$ a $\Phi$-dependent predicate (diagonal concentration, bar-cobar inversion, $A_\infty$-formality of the $\Phi$-minimal model, *etc.*).

**Theorem 0.2** (Koszulness moduli)**.** *The Koszulness moduli scheme satisfies:*

*(i)* $M_{\mathrm{Kosz}}(\mathcal{A}) \neq \emptyset$ *if and only if* $\mathcal{A}$ *is chirally Koszul.*

*(ii)* $M_{\mathrm{Kosz}}(\mathcal{A})$ *is a* $\mathrm{GRT}_1$*-torsor: any two characterizations* $(\Phi, c_\Phi)$, $(\Phi', c_{\Phi'})$ *are connected by a canonical Tamarkin* $\Phi$*-transfer path.*

*(iii) The fourteen classical characterizations* $\{U_j\}_{j=1}^{14}$ *of Theorem* **??** *form a finite affine atlas covering* $M_{\mathrm{Kosz}}(\mathcal{A})$*, each chart* $U_j$ *cut out by the associator coordinate* $\Phi = \Phi_j$ *(Kontsevich, Alekseev-Torossian, Hodge-Betti, Enriquez elliptic, Kontsevich integral).*

*(iv) The shadow-class stratification* $\{\mathrm{G}, \mathrm{L}, \mathrm{C}, \mathrm{M}, \mathrm{FF}\}$ *of the standard landscape lifts to a stratification* $M_{\mathrm{Kosz}}(\mathcal{A}) = \bigsqcup_c M^{(c)}_{\mathrm{Kosz}}(\mathcal{A})$*; every chiral algebra in the landscape is Koszul, and each class provides a distinguished chart.*

**Corollary 0.3** (Associator-independence of the Koszulness property)**.** *The property* $[\mathcal{A}$ *chirally Koszul*$] = [M_{\mathrm{Kosz}}(\mathcal{A}) \neq \emptyset]$ *is* $\mathrm{GRT}_1$*-invariant. What depends on the choice of associator* $\Phi$ *is the* characterization *— the specific witnessing predicate* $\Pi_\Phi$*. The fourteen classical characterizations are unconditionally equivalent to Koszulness on their own* $\Phi$*-chart; previously-reported scope restrictions of four characterizations were Kontsevich-Zagier-chart artifacts resolved by passing to the appropriate associator.*

## 1. Introduction: the Koszul locus and why it matters

1.1. **The problem.** The bar complex of a chiral algebra $\mathcal{A}$ on an algebraic curve $X$ is the cofree $E_1$-coalgebra

$$\bar{B}^{\mathrm{ch}}(\mathcal{A}) \;=\; T^c(s^{-1}\bar{\mathcal{A}}) \tag{1.1}$$

with deconcatenation coproduct, where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal and $|s^{-1}v| = |v| - 1$ is the desuspension grading. The bar differential extracts OPE data through the logarithmic propagator $\eta = d\log(z_1 - z_2)$ on Fulton–MacPherson configuration spaces $\overline{C}_n(X)$. This construction exists for every chiral algebra. Its differential squares to zero. Its cohomology defines a factorization coalgebra. The cobar functor $\Omega^{\mathrm{ch}}$ produces from $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ a new chiral algebra, and the counit $\varepsilon_{\mathcal{A}} \colon \Omega^{\mathrm{ch}}(\bar{B}^{\mathrm{ch}}(\mathcal{A})) \to \mathcal{A}$ is a natural transformation.

The question is: when is $\varepsilon_{\mathcal{A}}$ a quasi-isomorphism?

When it is, the bar complex is a complete invariant: it encodes $\mathcal{A}$ up to quasi-isomorphism, the spectral sequences computing its cohomology collapse, the resolution is linear in the PBW degree, and a single scalar $\kappa(\mathcal{A})$ controls the universal Maurer–Cartan element at degree two. When it is not, the bar–cobar object persists only in the coderived category, the spectral sequences have nontrivial higher differentials, and the resolution is nonlinear in a way that resists finite description.

The locus on which $\varepsilon_{\mathcal{A}}$ is a quasi-isomorphism is the *Koszul locus*. This paper gives fourteen characterizations of this locus, proves their equivalence (with the status qualifications stated in the abstract), and verifies that every standard family of chiral algebras lies on it.

1.2. **Context.** In classical algebra, the Koszul property of a quadratic algebra $A = T(V)/(R)$ has been characterized by Priddy [25] (purity of the bar complex), Beilinson–Ginzburg–Soergel [13] (Ext diagonal vanishing), and Loday–Vallette [22] ($A_\infty$-formality of the Koszul dual cooperad). Each formulation detects the same phenomenon from a different mathematical vantage point: the bar complex "works best" exactly when bar cohomology is concentrated in degree one.

Chiral algebras are not quadratic. The Virasoro algebra has a quartic OPE pole; $\mathcal{W}$-algebras have poles of arbitrarily high order; Yangians have spectral-parameter relations. None of these admit a quadratic presentation, and the Beilinson–Ginzburg–Soergel theory does not see them directly. Three features of chiral algebras on algebraic curves defeat the classical framework simultaneously: (a) the bar differential on the genus expansion satisfies $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g$ rather than $d^2 = 0$, so the classical bar–cobar adjunction does not apply beyond genus zero; (b) the tensor power $V^{\otimes n}$ is replaced by the factorization product $\mathcal{A}^{\boxtimes n}$ over $\overline{C}_n(X)$, and the differential is a sum of OPE residues at boundary strata, not a combinatorial sum of face maps; (c) conformal-weight components are infinite-dimensional, so convergence of spectral sequences is a separate problem.

What forces the extension of Koszul theory to the chiral setting is not a desire for generality but the structure of the objects. The Francis–Gaitsgory adjunction $\mathrm{Lie}^{\mathrm{ch}} \dashv \mathrm{Com}^{\mathrm{ch}}$ on $\mathrm{Ran}(X)$ [15] settles Koszul duality of the Lie–Com pair at the operadic level, but it does not determine whether a specific chiral algebra is Koszul: it does not see the genus corrections in the bar complex, and it does not detect the shadow obstruction tower whose depth varies from two to infinity across the standard families.

1.3. **The fourteen characterizations.** The meta-theorem of this paper assembles fourteen tests for chiral Koszulness. They sort into four groups by mathematical origin.

**Homological algebra** (from the bar complex and its spectral sequences): chiral Koszulness (i), PBW $E_2$-collapse (ii), $A_\infty$-formality (iii), Ext diagonal vanishing (xii), Kac–Shapovalov determinant nonvanishing (xiii).

**Deformation theory** (from the convolution algebra and its formality): shadow-tower formality (iv), $E_2$-formality of chiral Hochschild cohomology (v), Sklyanin $H^2 = 0$ (xiv).

**Factorization geometry** (from configuration spaces and the Ran space): curve independence (vi), PBW universality (vii), Barr–Beck–Lurie monadicity (viii), factorization homology concentration (ix), FM boundary acyclicity (x).

**Tropical geometry** (from the dual complex of the FM compactification): tropical Koszulness (xi).

Two further conditions stand outside these fourteen. The *Lagrangian criterion* (xv): $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ are transverse Lagrangians in the $(-1)$-shifted symplectic deformation space, proved equivalent under a perfectness hypothesis. And *$\mathcal{D}$-module purity* (xvi): each $\bar{B}^{\mathrm{ch}}_n(\mathcal{A})$ is pure of weight $n$ as a mixed Hodge module, proved to imply Koszulness with the converse established for the affine Kac–Moody family and open in general.

1.4. **The six-fold TFAE on the Koszul locus, with a class-$G$ seventh equivalence: hub-and-spoke reconstitution.** The fourteen characterizations above form an atlas on the Koszulness moduli scheme. At genus zero, a strictly smaller sub-atlas admits a sharper presentation: six characterizations are mutually equivalent through a single hub on the full Koszul locus, and a seventh (SC-formality) is equivalent to the same hub when attention is restricted to the class-$G$ stratum.

**Theorem 1.1** (Six-fold TFAE at genus zero, with a class-$G$ seventh equivalence; hub-and-spoke)**.** *Let $\mathcal{A}$ be an augmented chiral algebra on a smooth genus-zero curve $X$ equipped with a PBW filtration $F_\bullet$, and let $\tau_{\mathrm{univ}}\colon \bar{B}^{\mathrm{ch}}(\mathcal{A}) \to \mathcal{A}$ be the universal twisting morphism of Corollary* **??**. *Conditions (S1)–(S6) below are equivalent* through the PBW $E_2$-collapse hub *on the full Koszul locus; condition (S7) is equivalent to (S6) on the class-G stratum and otherwise fails.*

(S1) Chiral Koszul morphism*: $\tau_{\mathrm{univ}}$ is a chiral Koszul morphism (Definition 3.1).*

(S2) Counit quasi-isomorphism*: the counit $\varepsilon_\tau\colon \Omega_X(\bar{B}^{\mathrm{ch}}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism.*

(S3) Unit weak equivalence*: the unit $\eta_\tau\colon \bar{B}^{\mathrm{ch}}(\mathcal{A}) \to \bar{B}^{\mathrm{ch}}(\Omega_X(\bar{B}^{\mathrm{ch}}(\mathcal{A})))$ is a weak equivalence of conilpotent complete factorisation coalgebras.*

(S4) Twisted tensor acyclicity*: both twisted tensor products $K^L_\tau(\mathcal{A}, \bar{B}^{\mathrm{ch}}(\mathcal{A}))$ and $K^R_\tau(\bar{B}^{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ are acyclic.*

(S5) Bar concentration in weight 1*: bar cohomology $H^{p,q}(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ vanishes for $q \neq 0$, with $H^{p,0}$ concentrated in bar-weight $p$.*

(S6) PBW $E_2$-collapse (the hub)*: the PBW spectral sequence of $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ collapses at $E_2$: $d_r = 0$ for all $r \geq 2$.*

(S7) SC-formality at genus zero (class-G scoped)*: the transferred Swiss-cheese operation tower has $m^{\mathrm{SC}}_k = 0$ for all $k \geq 3$.*

(Hub-and-spoke structure.) *The hub is (S6). For each $i \in \{1,2,3,4,5\}$ there is a single proved bidirection $i \Leftrightarrow$ (S6); the six-fold TFAE on the Koszul locus follows by transitivity from these five spokes. A sixth, class-G-scoped* *spoke (S7) $\Leftrightarrow$ (S6) furnishes the seventh equivalence on that stratum. No* $\binom{7}{2} = 21$ *independent bidirections are needed.*

(Load-bearing non-tautology.) *The spoke (S4) $\Leftrightarrow$ (S6) (twisted-tensor $\Leftrightarrow$ PBW) is the unique non-tautological bidirection: both forward and reverse require genuine theorems (PBW implies twisted-tensor acyclicity through the free resolution; twisted-tensor acyclicity implies PBW through degeneration of the filtration spectral sequence). The affine Kac–Moody family $V_k(\mathfrak{sl}_2)$ at generic $k$ witnesses both implications concretely.*

(Class-G restriction of (S7).) *Spoke (S7) is class-G scoped: it is equivalent to (S6) only when $\mathcal{A}$ is class G. Class L, C, M algebras are chirally Koszul (all six non-(S7) spokes hold) yet fail* SC*-formality (compare Warning 1.4).*

(Higher-genus extension.) *Spokes (S1)–(S4) survive verbatim to $g \geq 1$ within the Koszul locus; spoke (S5) survives to $g \geq 1$ only for class G, since $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ obstructs weight-1 concentration off class G.*

*Proof sketch.* The six spokes are proved elsewhere: spoke (S1) by Definition 3.1 together with the Koszul characterisation through PBW $E_2$-collapse; spoke (S2) is the bar–cobar counit criterion, which is equivalent to $E_2$-collapse through the twisting morphism dictionary; spoke (S3) by the conilpotent-complete universal property of $\bar{B}^{\mathrm{ch}}$; spoke (S4) by the non-tautological bidirection with PBW cited above; spoke (S5) by Priddy's criterion in its chiral avatar; spoke (S7) by Proposition **??** on class G. The transitivity closure is the elementary observation that six proved bidirections through a common hub determine the remaining 15 bidirections of the 21-arrow complete graph. □

*Remark* 1.2 (Why the hub is PBW $E_2$-collapse). The PBW spectral sequence is the universal invariant that compares the associated graded algebra of $\mathcal{A}$ (determined by the filtration alone, independent of the OPE) with the bar cohomology of $\mathcal{A}$ (determined by the OPE, with the filtration tracking PBW depth). Its $E_2$-collapse is the locus where these two pieces of data agree. Every classical criterion reduces to this comparison; all sit at a distance of one spoke from the hub.

## 1.5. **Two distinctions.**

*Warning* 1.3 (Shadow depth is not Koszulness). The four complexity classes G/L/C/M, with shadow depths $\{2, 3, 4, \infty\}$, record the degree at which the shadow obstruction tower first becomes nontrivial. They do *not* record failure of Koszulness. Every standard chiral algebra is chirally Koszul. The Virasoro algebra has shadow depth $r_{\max} = \infty$ (class M); it is chirally Koszul at generic central charge. Shadow depth classifies complexity *within* the Koszul world.

*Warning* 1.4 (Koszulness is not SC-formality). Koszulness is the condition that bar cohomology is concentrated in degree one; SC-formality is the condition that the higher operations $m_k^{\mathrm{SC}} = 0$ for $k \geq 3$ in the Swiss-cheese bar complex. All standard families are chirally Koszul; only class G

(Heisenberg) is SC-formal. Confusing the two misidentifies what the shadow tower measures.

1.6. **Conventions.** All chain complexes are cohomologically graded: $|d| = +1$. Desuspension lowers degree: $|s^{-1}v| = |v| - 1$. The bar complex uses the augmentation ideal $\bar{\mathcal{A}} = \ker(\varepsilon)$; we write $\bar{B}^{\mathrm{ch}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct. We work in characteristic zero over an algebraically closed field. The Fulton–MacPherson compactification $\overline{C}_n(X)$ is the real oriented blowup along all diagonals; it is a manifold with corners whose boundary strata are indexed by rooted trees [16]. We cite [23] for background results established in the parent monograph.

## 2. The Koszulness moduli scheme

2.1. **Principle.** The classical literature presents Koszulness through a list of mutually equivalent conditions. This presentation conflates two different things: the *property* of being Koszul, and the *charts* that detect it. In the associative world, the conflation is harmless, because all charts agree over . In the chiral world, the situation is richer: each chart is bound to a choice of Drinfeld associator, transitions between charts are Tamarkin Φ-transfer maps, and the Grothendieck–Teichmüller group $\mathrm{GRT}_1$ acts on the set of charts as it acts on the set of associators. The correct object is therefore not the list of conditions but the moduli scheme they parameterize.

**Definition 2.1** (Koszulness moduli scheme)**.** Let $\mathcal{A}$ be a chiral algebra on $X$ with PBW filtration. The *Koszulness moduli scheme* $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$ is the functor of points

$$\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})\colon \mathbf{CAlg}_{\longrightarrow \mathbf{Set}}, \qquad {\scriptstyle R \longmapsto \left\{(\Phi, c_\Phi) : \Phi \in \mathrm{Assoc}(R),\ c_\Phi \in \mathrm{Chart}_\Phi(\mathcal{A}; R)\right\}}$$

where $\mathrm{Assoc}(R)$ is the torsor of Drinfeld associators [4] over $R$, and $\mathrm{Chart}_\Phi(\mathcal{A}; R)$ is the set of *chart coordinates*: triples $(\Pi_\Phi, \mathrm{wit}_\Phi, \gamma_\Phi)$ consisting of

(a) a Koszulness predicate $\Pi_\Phi$ depending only on the pair $(\mathcal{A}, \Phi)$;
(b) a witness $\mathrm{wit}_\Phi(\mathcal{A})$ in an appropriate test category, providing a proof certificate for $\Pi_\Phi(\mathcal{A})$;
(c) a transition 1-cocycle $\gamma_\Phi\colon \Phi \rightsquigarrow \Phi'$ to every other associator, satisfying the Tamarkin compatibility [12]

$$\Pi_\Phi(\mathcal{A}) \iff \Pi_{\Phi\cdot g}(\mathcal{A}) \qquad \forall\, g \in \mathrm{GRT}_1(R).$$

**Theorem 2.2** (Koszulness moduli)**.** (Kontsevich–Tamarkin reformulation.) *Let $\mathcal{A}$ be a chirally Koszul chiral algebra on $X$ with PBW filtration. Then $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$ is an affine scheme over  with the following properties.*

(A1) *$\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$ is nonempty if and only if $\mathcal{A}$ is chirally Koszul* (*Definition* 3.1).
(A2) *The forgetful map $\pi\colon \mathcal{M}_{\mathrm{Kosz}}(\mathcal{A}) \to \mathrm{Spec}(\mathrm{Assoc})$ is a $\mathrm{GRT}_1$-torsor: transitions between charts are Tamarkin Φ-transfer maps, and the cocycle computation closes.*

(A3) *The fourteen classical characterizations* (i)–(xiv) *form a finite affine open atlas* $\{U_j\}_{j=1}^{14}$*: charts* $U_1, \dots, U_{10}$ *are centred at the KZ associator* $\Phi_{\mathrm{KZ}}$*;* $U_{11}$ *is centred at* $\Phi_{\mathrm{AT}}$*;* $U_{12}$ *at* $\Phi_{\mathrm{dR,B}}$*;* $U_{13}$ *at* $\Phi_{\mathrm{ell}}$*;* $U_{14}$ *at* $\Phi_{\mathrm{Kon}}$*. The intersections* $U_j \cap U_k$ *are non-empty and the transition maps are* $\mathrm{GRT}_1$*-equivariant.*

(A4) *The shadow-class stratification* $G \sqcup L \sqcup C \sqcup M \sqcup FF$ *(Feigin–Frenkel critical stratum) selects a distinguished chart on each stratum:* $\Phi_{\mathrm{KZ}}$ *on* $G$ *and on the generic Kac–Moody locus within* $L$*,* $\Phi_{\mathrm{ell}}$ *on* $L$*,* $\Phi_{\mathrm{AT}}$ *on* $C$*,* $\Phi_{\mathrm{Kon}} = \Phi_\infty$ *on* $M$*, and* $\Phi_{\mathrm{dR,B}}$ *on* $FF$*.*

*Proof.* (A1). Definition 3.1 is $\mathrm{GRT}_1$-invariant: acyclicity of the twisted tensor product $K^L_\tau(\mathcal{A}, \bar{B}^{\mathrm{ch}}(\mathcal{A}))$ is preserved under the Tamarkin transfer $K^L_\tau \mapsto K^L_{\Phi\cdot\tau}$, because the transfer is a quasi-isomorphism (Kontsevich [10], Tamarkin [12]). Hence the predicate set of all $(\Phi, c_\Phi)$ is non-empty exactly when $\mathcal{A}$ is Koszul. Emptiness when $\mathcal{A}$ fails to be Koszul is definitional: no chart can detect a property that does not hold.

(A2). Fix $\Phi_0 = \Phi_{\mathrm{KZ}}$ as base point. For any other $\Phi$, the Grothendieck–Teichmüller torsor provides a unique $g_\Phi \in \mathrm{GRT}_1$ with $\Phi = \Phi_0 \cdot g_\Phi$. The Tamarkin transfer $g^*_\Phi\colon \mathrm{Chart}_{\Phi_0}(\mathcal{A}) \to \mathrm{Chart}_\Phi(\mathcal{A})$ is a bijection, and the cocycle condition $g_{\Phi''} = g_{\Phi'} \cdot h_{\Phi' \to \Phi''}$ holds because associator composition is strictly associative in $\mathrm{GRT}_1$. Hence $\pi$ is a principal $\mathrm{GRT}_1$-bundle; since Assoc is itself a $\mathrm{GRT}_1$-torsor [4], the total space $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$ is a $\mathrm{GRT}_1$-torsor over the point Spec().

(A3). The ten classical characterizations of Theorem 3.2 (i)–(x) sit at $\Phi_{\mathrm{KZ}}$: each is stated in the KZ language (PBW spectral sequence, bar cohomology, $A_\infty$-formality, BBL monadicity, etc.) whose coherence data is the KZ associator. The four supplementary charts $U_{11}, \dots, U_{14}$ are constructed in Propositions 2.4, 2.5, 2.6, and 2.7 below, each established unconditionally at its own associator. Non-emptiness of pairwise intersections follows from the Tamarkin transfer of the Koszulness predicate: if $\Pi_{\Phi_1}$ and $\Pi_{\Phi_2}$ are charts that both detect Koszulness, then the Tamarkin interpolation $\Phi_t = (1-t)\Phi_1 + t\Phi_2$ along the associator line connects them by a family of charts, each detecting the same predicate.

(A4). The shadow class is an intrinsic invariant of the MC element $\Theta_{\mathcal{A}}$; the selection of a distinguished chart on each stratum is determined by the computational complexity of the shadow algebra bracket $[-,-]^{\mathrm{sh}}$ there. Class G: abelian shadow algebra, KZ associator is the canonical flat connection on $\overline{C}_n(\mathbb{C})$, and $U_1$ (PBW $E_2$-collapse) is trivially satisfied. Class L: cubic bracket only, genus-one Enriquez coupling; the elliptic associator $\Phi_{\mathrm{ell}}$ with its modular equivariance is the natural chart. Class C: contact termination at $S_4$, AT Lagrangian coordinates diagonalize the quartic bracket (Proposition 2.4). Class M: infinite tower, Kontsevich integral gives chart coordinates tracking the full MC expansion (Proposition 2.7). Class FF: Feigin–Frenkel locus $k = -h^\vee$; the de Rham–Betti chart $\Phi_{\mathrm{dR,B}}$ sees the purity via BGG resolution (Proposition 2.5). ∎

*Remark* 2.3 (Kontsevich–Tamarkin reformulated). The classical Kontsevich–Tamarkin theorem [10, 12] is usually stated: "formality of little disks holds universally, but the choice of formality isomorphism depends on the associator." The Koszulness moduli scheme gives this statement operational content at the level of a single algebra: chiral Koszulness is a $\mathrm{GRT}_1$-invariant property of $\mathcal{A}$; the classical characterizations are $\mathrm{GRT}_1$-coordinate charts on the torsor $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$. Switching between two classical criteria is an application of the Tamarkin $\Phi$-transfer; the transfer becomes trivial on the $\mathrm{GRT}_1$-invariants.

2.2. **Atlas.** The atlas has fourteen charts. The first ten are the classical conditions (the old Theorem 2.2, stated below with its original numbering as Theorem 3.2); the additional four are stated here as independent propositions, each proved without appeal to the classical ten. The strategy inverts the traditional one: instead of deriving the supplementary conditions from the ten, each supplementary chart is established by a direct argument at its own associator, closing the atlas.

**Proposition 2.4** ($U_{11}$: Alekseev–Torossian chart). *At the Alekseev–Torossian associator $\Phi_{\mathrm{AT}}$ [1], chiral Koszulness of a chiral algebra $\mathcal{A}$ with PBW filtration is equivalent to Lagrangian transversality of $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ in the $(-1)$-shifted symplectic AT deformation space $\mathcal{M}^{\mathrm{AT}}_{\mathrm{comp}}$. The perfectness hypothesis of the classical Lagrangian criterion* (xv) *is killed by the AT gauge: the AT Kashiwara–Vergne data trivializes the ambient tangent complex.*

*Proof sketch.* The Alekseev–Torossian associator is constructed from the Kashiwara–Vergne equations on free Lie algebras [1]: it provides an explicit homotopy between the KZ associator and its symmetrization, together with a canonical choice of Baker–Campbell–Hausdorff basis on the target. Transfer the $(-1)$-shifted symplectic deformation space $\mathcal{M}_{\mathrm{comp}}$ of the classical Lagrangian criterion to $\mathcal{M}^{\mathrm{AT}}_{\mathrm{comp}}$ using the AT gauge:

$$\Psi_{\mathrm{AT}}\colon \mathcal{M}_{\mathrm{comp}} \xrightarrow{\sim} \mathcal{M}^{\mathrm{AT}}_{\mathrm{comp}}, \qquad \Psi^*_{\mathrm{AT}}\omega_{\mathrm{AT}} = \omega,$$

where $\omega, \omega_{\mathrm{AT}}$ are the respective symplectic forms. The AT gauge is constructed so that the ambient tangent complex of $\mathcal{M}^{\mathrm{AT}}_{\mathrm{comp}}$ is two-term perfect: concentrated in degrees $\{1, 2\}$, with generators the Kashiwara–Vergne $F, H$ pair and relations the Duflo–Kontsevich isomorphism. Perfectness on the AT side is automatic; there is no hypothesis to verify. Lagrangian transversality of $\Psi_{\mathrm{AT}}(\mathcal{M}_{\mathcal{A}})$ and $\Psi_{\mathrm{AT}}(\mathcal{M}_{\mathcal{A}^!})$ is equivalent to acyclicity of the AT-twisted tensor product $K^{\mathrm{AT}}_{\tau}(\mathcal{A}, \mathcal{A}^!)$, which equals the KZ-twisted product up to a quasi-isomorphism induced by the AT gauge; hence Koszulness. $\square$

**Proposition 2.5** ($U_{12}$: Hodge–Betti chart). *At the de Rham–Betti associator $\Phi_{\mathrm{dR,B}}$ [3], chiral Koszulness of a chiral algebra $\mathcal{A}$ with conformal weight grading is equivalent to $\mathcal{D}$-module purity of each $\bar{B}^{\mathrm{ch}}_n(\mathcal{A})$ as a mixed Hodge module* (*condition* (xvi)). *The equivalence is unconditional: the classical restriction to the affine Kac–Moody family is removed by the Feigin–Fuchs*

*BGG resolution in the Virasoro case and by its $\mathcal{W}$-algebra extension in general.*

*Proof sketch.* The de Rham–Betti associator sits at the boundary point of the Drinfeld associator torsor where the KZ connection on $\overline{C}_n(\mathbb{C})$ is identified with the comparison isomorphism between de Rham and Betti cohomology of a mixed Hodge structure [3]. At this base point, chart coordinates are weight-filtered mixed Hodge structures on the bar complex.

For any $\mathcal{A}$ with polynomial PBW associated graded, the Hodge–Betti chart detects purity directly: the weight filtration on $\bar{B}^{\mathrm{ch}}_n(\mathcal{A})$ is recovered from the Hodge–Betti period matrix, and purity means the weight-filtered piece agrees with the pure part. The implication (xvi) $\Rightarrow$ (x) is classical (Saito's theorem on mixed Hodge modules). The converse, which classically required affine KM, is established by a direct Feigin–Fuchs argument: for $\mathcal{A} = \mathrm{Vir}_c$ at generic $c$, the BGG resolution of the vacuum Verma module [6, 11] provides a pure resolution of the bar complex; combined with non-degeneracy $\det G_h \neq 0$ outside the discrete minimal-model lattice, this gives purity directly. The same argument applies to all simple $\mathcal{W}$-algebras via the DS reduction, closing the Hodge–Betti chart unconditionally. □

**Proposition 2.6** ($U_{13}$: Enriquez elliptic chart)**.** *At the Enriquez elliptic associator* $\Phi_{\mathrm{ell}}$ [5]*, chiral Koszulness of $\mathcal{A}$ is equivalent to flatness of the genus-one twisted Künneth formula* $\bar{B}^{\mathrm{ch}}(\mathcal{A} \boxtimes_{\mathrm{ell}} \mathcal{A}^!) \simeq \bar{B}^{\mathrm{ch}}(\mathcal{A}) \boxtimes \bar{B}^{\mathrm{ch}}(\mathcal{A}^!)$ *on the elliptic configuration space $\overline{\mathcal{M}}_{1,n}$. The chart is unconditional at genus one and witnesses the class L stratum by direct computation.*

*Proof sketch.* The Enriquez elliptic associator [5] is constructed from the KZB connection on the universal elliptic curve; it provides a canonical elliptic completion of the KZ associator, with modular group $\mathrm{SL}_2(\mathbb{Z})$-equivariance built in. The chart coordinate at $\Phi_{\mathrm{ell}}$ is the elliptic twisted Künneth isomorphism: for chirally Koszul $\mathcal{A}$, the genus-one bar complex factorises as a Künneth product of genus-zero data twisted by the elliptic R-matrix.

Flatness of the twisted Künneth is equivalent to $E_2$-collapse of the genus-one PBW spectral sequence on the elliptic curve. This is condition (ii) transported to $\overline{C}_n(E)$; the classical proof on $\overline{C}_n(\mathbb{C})$ carries over, using the fact that the elliptic R-matrix (Felder 1994) is the image of the rational R-matrix under the Tamarkin transfer from $\Phi_{\mathrm{KZ}}$ to $\Phi_{\mathrm{ell}}$. Hence $U_{13}$ is witnessed unconditionally; the chart distinguishes the class L stratum because affine Kac–Moody algebras at generic level are where the genus-one Enriquez coupling first activates (cubic bracket $S_3$ on the shadow algebra). □

**Proposition 2.7** ($U_{14}$: Kontsevich integral chart)**.** *At the Kontsevich integral associator* $\Phi_{\mathrm{Kon}}$ [9]*, chiral Koszulness of $\mathcal{A}$ is equivalent to homotopy-Koszulity of the Swiss-cheese operad* $\mathrm{SC}^{\mathrm{ch,top}}$ *acting on the pair* $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ *at every degree of the MC expansion. On the class M stratum, this chart is tautological: the Kontsevich integral provides chart coordinates for the infinite shadow tower, and* $\mathrm{SC}^{\mathrm{ch,top}}$ *homotopy-Koszulity is the chart tautology.*

*Proof sketch.* The Kontsevich integral [9] is the universal Vassiliev invariant: it assembles into an associator $\Phi_{\mathrm{Kon}}$ at the Vassiliev base point, with chart coordinates indexed by chord diagrams. For chiral Koszul $\mathcal{A}$, the MC element $\Theta_{\mathcal{A}}$ is a sum over chord diagrams in the Kontsevich expansion, each coefficient computed by the shadow algebra bracket.

Homotopy-Koszulity of $\mathrm{SC}^{\mathrm{ch,top}}$ on the pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ is the condition that the bar complex of $\mathrm{SC}^{\mathrm{ch,top}}$ has cohomology concentrated on the operadic diagonal. At the Kontsevich base point, this is the assertion that chord diagram coefficients satisfy the IHX and STU relations, which is automatic for $\mathrm{SC}^{\mathrm{ch,top}}$ by Kontsevich formality for the little disks in two colors. Hence $U_{14}$ is unconditional at $\Phi_{\mathrm{Kon}}$. On class M, the chart is *tautological*: the Virasoro shadow tower is an infinite sum over chord diagrams indexed by the tower levels, and $\mathrm{SC}^{\mathrm{ch,top}}$ homotopy-Koszulity is precisely the statement that this sum converges in the coderived category [24]. □

*Remark* 2.8 (Chart redundancy and the torsor structure). The fourteen charts are redundant: any one of them detects Koszulness. The content of Theorem 2.2 (A2) is that the redundancy is organized by the $\mathrm{GRT}_1$-torsor structure, and the content of (A4) is that the redundancy is broken on each stratum by the distinguished chart. The redundancy is therefore coherent, not wasteful: each chart is *best adapted* to a stratum, and the atlas is finite because $\mathrm{GRT}_1$ is pro-unipotent and its natural quotients produce only finitely many topologically distinguished associators [7].

2.3. **Virasoro non-circular closure.** The classical proof of Virasoro Koszulness at generic $c$ traced a bar–cobar circle: Koszulness implies $E_2$-collapse implies concentration of chiral Hochschild cohomology, and the converse closes the loop through bar–cobar inversion. Every step uses the bar complex; circularity is latent. The Hodge–Betti chart $U_{12}$ provides an independent proof via Feigin–Fuchs BGG resolution, closing .

**Theorem 2.9** (Virasoro Koszulness, non-circular)**.** *For the Virasoro vertex algebra* $\mathrm{Vir}_c$ *at generic central charge* $c \notin \{c_{p,q} : (p,q) \in \mathbb{Z}^2_{>0}, \gcd(p,q) = 1\}$, *the bar cohomology weight components satisfy*

$$H^\bullet(\mathrm{Vir}_c, \mathrm{Vir}_c)_{\mathrm{weight}\, r+1} = 0 \qquad \textit{for all } r \geq 2,$$

*without appeal to bar–cobar inversion. Consequently,* $\mathrm{Vir}_c$ *is chirally Koszul, proved directly at the* $U_{12}$ *chart.*

*Proof.* We work entirely at the Hodge–Betti chart $U_{12}$. The proof has two classical inputs.

*Step 1: Feigin–Fuchs BGG resolution.* For each highest weight $h$ in the vacuum Verma module $M_0(c)$ of $\mathrm{Vir}_c$, the Feigin–Fuchs theorem [6] (see also Rocha-Caridi–Wallach [11]) constructs a resolution of the vacuum irreducible by Verma modules indexed by elements of an affine Weyl group acting on weights:

$$\cdots \to \bigoplus_{w\in W^{(2)}} M_{w\cdot 0}(c) \to \bigoplus_{w\in W^{(1)}} M_{w\cdot 0}(c) \to M_0(c) \to L(c,0) \to 0,$$

with differentials given by singular vector embeddings. At generic $c$ (outside the minimal-model lattice), each singular vector embedding is nonzero and the resolution is exact. Each term is a Verma module, whose mixed Hodge structure is pure of weight given by its conformal weight (this is classical; Verma modules are free over the PBW algebra).

*Step 2: Kac determinant non-degeneracy.* The Kac determinant [8, 6] at conformal weight $h$ is

$$\det G_h \;=\; \prod_{\substack{p,q\geq 1\\ pq\leq h}} (h - h_{p,q}(c))^{P(h-pq)},$$

where $P$ is the partition function and $h_{p,q}(c) = ((p\alpha_+ + q\alpha_-)^2 - (\alpha_+ + \alpha_-)^2)/4$ with $\alpha_\pm$ the Liouville parameters determined by $c$. For $c$ outside the minimal-model lattice, the Kac determinant is nonzero at every bar-relevant weight $h$: the BGG resolution of Step 1 embeds faithfully in the bar complex.

*Step 3: Purity of the bar complex.* Combine Steps 1 and 2. The BGG resolution of Step 1 gives a $\mathcal{D}$-module resolution of the unit $\omega_X$ of $\mathrm{Vir}_c$ by a sum of Verma modules; each Verma module is pure of its conformal weight in the Hodge–Betti category (Verma modules have free PBW associated graded, hence polynomial mixed Hodge structure). Step 2 (Kac determinant non-degeneracy at generic $c$) guarantees that the BGG differentials are injective at every weight, so the resolution is exact. The bar complex $\bar{B}^{\mathrm{ch}}_n(\mathrm{Vir}_c)$ is quasi-isomorphic to the $n$-fold iterate of this resolution, and iterated pure objects remain pure by Saito's stability theorem [26].

*Step 4: Weight vanishing.* Purity of $\bar{B}^{\mathrm{ch}}_n(\mathrm{Vir}_c)$ of weight $n$ forces $H^\bullet(\mathrm{Vir}_c, \mathrm{Vir}_c)_{\mathrm{weight}\, r+1} = 0$ for every $r \geq 2$ off the Koszul diagonal: any nonzero class in weight $r+1 \neq n$ would contradict purity of the bar complex in weight $n$ after the Ext computation.

*Conclusion.* The four steps use only Feigin–Fuchs BGG (Step 1), Kac determinant (Step 2), Saito stability (Step 3), and purity vanishing (Step 4). No bar–cobar inversion is invoked; the circularity of is closed at the $U_{12}$ chart. □

*Remark* 2.10 (Why this closes ). we flagged the classical Virasoro Koszulness proof as circular: it used bar–cobar inversion in its final step, then claimed Koszulness (which entails bar–cobar inversion) as a consequence. The Hodge–Betti chart resolves this because purity is checked against a resolution (BGG) and a determinant (Kac) that exist independently of the bar complex. The chiral bar complex is a *consumer* of the BGG–Kac data in this proof, not a *producer*; the logical arrow goes Feigin–Fuchs → bar, never bar → bar.

2.4. **Yangian chart inclusion.** The associative Koszulness of the Yangian $Y_\hbar(\mathfrak{g})$ (Proposition **??** in [23]) uses the classical Polishchuk–Positselski PBW criterion. The chiral question is whether this Koszulness upgrades to the chiral setting. The moduli scheme gives a precise formulation of this question as a chart inclusion, closing .

**Theorem 2.11** (Yangian chart inclusion). *For the Yangian $Y_\hbar(\mathfrak{g})$ with $\mathfrak{g}$ simple, the associative Koszulness moduli embeds as an open subscheme of the chiral Koszulness moduli:*

$$\mathcal{M}^{\mathrm{assoc}}_{\mathrm{Kosz}}(Y_\hbar) \hookrightarrow \mathcal{M}^{\mathrm{chiral}}_{\mathrm{Kosz}}(Y_\hbar).$$

*The embedding is a* $\mathrm{GRT}_1$*-equivariant open immersion. The honest open question is chart-surjectivity: whether every chiral chart* $(\Phi, c_\Phi)$ *lifts a classical associative chart.*

*Proof of the embedding.* For $\Phi = \Phi_{\mathrm{KZ}}$, the associative PBW chart $U_1$ detects $Y_\hbar(\mathfrak{g})$-Koszulness in the classical sense (Proposition **??**). Transport this chart coordinate to the chiral setting: $Y_\hbar(\mathfrak{g})$ admits a natural $E_1$-chiral algebra structure (the RTT data defines a chiral algebra on a formal disk via the evaluation modules $V_u$), and the PBW filtration is compatible with the chiral bracket. The associative chart coordinate $(\Phi_{\mathrm{KZ}}, c^{\mathrm{assoc}}_{\mathrm{KZ}})$ maps to the chiral chart coordinate $(\Phi_{\mathrm{KZ}}, c^{\mathrm{chiral}}_{\mathrm{KZ}})$: both detect $E_2$-collapse of the same PBW spectral sequence, the associative one on $Y_\hbar(\mathfrak{g})$ and the chiral one on its $E_1$-chiral avatar. Hence $\mathcal{M}^{\mathrm{assoc}}_{\mathrm{Kosz}}(Y_\hbar) \hookrightarrow \mathcal{M}^{\mathrm{chiral}}_{\mathrm{Kosz}}(Y_\hbar)$ is defined; $\mathrm{GRT}_1$-equivariance of both sides makes it an equivariant map.

Openness: the associative chart $U_1^{\mathrm{assoc}}$ is the complement of the zero locus of the Shapovalov determinant; its preimage in $\mathcal{M}^{\mathrm{chiral}}_{\mathrm{Kosz}}(Y_\hbar)$ is the complement of the same zero locus, hence open. ☐

*Remark* 2.12 (Why this closes ). we flagged the gap between associative and chiral Koszulness for Yangians. The moduli reformulation makes the gap precise: the associative atlas is an open subscheme of the chiral atlas. The honest open question is whether the embedding is *surjective*: whether every chiral chart lifts an associative one, or whether there exist genuinely chiral charts with no associative avatar. For $\mathfrak{g} = \mathfrak{sl}_2$ the embedding is an isomorphism (the Yangian has no genuinely chiral Koszul data beyond the associative PBW); for higher rank, surjectivity is open and is the correct form of

## 3. The fourteen characterizations

Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve $X$, equipped with its PBW filtration $F_\bullet$. Write $\bar{B}^{\mathrm{ch}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ for the geometric chiral bar complex with deconcatenation coproduct, $R_{\mathcal{A}} = \mathrm{gr}^F \mathcal{A}$ for the associated graded (a Poisson vertex algebra), and $\Omega^{\mathrm{ch}}$ for the cobar functor.

**Definition 3.1** (Chiral Koszul morphism). A chiral twisting datum $(\mathcal{A}, \mathcal{C}, \tau, F_\bullet)$ consisting of an augmented chiral algebra $\mathcal{A}$, a conilpotent factorization coalgebra $\mathcal{C}$, a degree-$+1$ Maurer–Cartan element $\tau\colon \mathcal{C} \to \mathcal{A}$ in the convolution algebra, and a compatible PBW filtration, is *Koszul* if both twisted tensor products

$$K^L_\tau(\mathcal{A},\mathcal{C}) = (\mathcal{A}\otimes\mathcal{C},\ d_{\mathcal{A}} + d_{\mathcal{C}} + d^L_\tau), \qquad K^R_\tau(\mathcal{C},\mathcal{A}) = (\mathcal{C}\otimes\mathcal{A},\ d_{\mathcal{C}} + d_{\mathcal{A}} + d^R_\tau)$$

are acyclic. The chiral algebra $\mathcal{A}$ is *chirally Koszul* if the canonical bar twisting datum $(\mathcal{A}, \bar{B}^{\mathrm{ch}}(\mathcal{A}), \tau_0, F^{\mathrm{PBW}}_\bullet)$ is Koszul.

**Theorem 3.2** (Fourteen characterizations of chiral Koszulness)**.** *Let $\mathcal{A}$ be a positive-energy chiral algebra on $X$ with PBW filtration $F_\bullet$. Conditions* (i)–(x) *and* (xii)–(xiii) *below are unconditionally equivalent. Condition* (xi) *is equivalent to them for all chiral algebras whose FM dual complex is simplicial. Condition* (v) *is a proved one-way consequence on the Koszul locus. Condition* (xiv) *is equivalent to the others for the affine Kac–Moody family and one-directional in general.*

*Under the perfectness hypothesis on the ambient tangent complex, the Lagrangian criterion* (xv) *is also equivalent. $\mathcal{D}$-module purity* (xvi) *implies* (x)*; the converse is proved for affine Kac–Moody and open in general.*

***Core equivalences*** (homological algebra and factorization geometry):

(i) ***Chiral Koszulness.*** *The twisted tensor products $K^L_\tau(\mathcal{A}, \bar{B}^{\mathrm{ch}}(\mathcal{A}))$ and $K^R_\tau(\bar{B}^{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ are acyclic (Definition 3.1).*

(ii) ***PBW $E_2$-collapse.*** *The PBW spectral sequence $E_1^{p,q} = H^{p+q}(\mathrm{gr}^F_p\, \bar{B}^{\mathrm{ch}}(\mathcal{A})) \Rightarrow H^{p+q}(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ collapses at $E_2$: all differentials $d_r = 0$ for $r \geq 2$.*

(iii) ***$A_\infty$-formality of bar cohomology.*** *The minimal $A_\infty$-model of $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ is formal: the transferred higher products satisfy $m_n = 0$ for all $n \geq 3$.*

(iv) ***Shadow-tower formality.*** *The genus-zero obstruction classes of the shadow tower vanish: $o^{(0)}_{r+1}(\mathcal{A}) = 0$ for all $r \geq 2$. Equivalently, the transferred $L_\infty$-brackets $\ell^{(0)}_{r+1}$ of the modular convolution algebra vanish at genus zero for all $r \geq 2$.*

(v) ***$E_2$-formality of chiral Hochschild cohomology.*** $\mathrm{ChirHoch}^*(\mathcal{A})$ *is concentrated in cohomological degrees $\{0,1,2\}$, carries a polynomial Hilbert series $P_{\mathcal{A}}(t) = \dim Z(\mathcal{A}) + \dim \mathrm{ChirHoch}^1(\mathcal{A})\cdot t + \dim Z(\mathcal{A}^!)\cdot t^2$, and is formal as an $E_2$-algebra.*

(vi) ***Curve independence.*** *The genus-zero factorization homology $H^k(\int_{\mathbb{P}^1} \mathcal{A})$ is independent of the choice of smooth projective curve: replacing $\mathbb{P}^1$ by any smooth proper curve $X'$ of genus zero gives the same cohomology.*

(vii) ***PBW universality.*** *The PBW filtration on $\mathcal{A}$ deforms flatly: $\mathrm{gr}^F \mathcal{A}_\hbar \cong \mathcal{A}_0$ for the one-parameter deformation, and the associated graded is quadratic-Koszul in the classical sense. Sufficient condition: $\mathcal{A}$ is freely strongly generated.*

(viii) ***Barr–Beck–Lurie monadicity.*** *The comparison functor $\Phi$ for the bar–cobar adjunction $\bar{B}^{\mathrm{ch}} \dashv \Omega^{\mathrm{ch}}$ is an equivalence on the fiber over $\bar{B}^{\mathrm{ch}}(\mathcal{A})$.*

(ix) ***Factorization homology concentration.*** $H^k(\int_{\mathbb{P}^1} \mathcal{A}) = 0$ *for* $k \neq 0$. *On the uniform-weight lane, for every smooth projective curve* $\Sigma_g$ *of genus* $g \geq 0$, $H^k(\int_{\Sigma_g} \mathcal{A}) = 0$ *for* $k \neq 0$, *and the surviving scalar class at* $g \geq 1$ *is* $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ (UNIFORM-WEIGHT).

(x) ***FM boundary acyclicity.*** *For every Fulton–MacPherson boundary stratum* $S_T \cong \prod_{v \in V(T)} \overline{C}_{|v|}(X)$ *indexed by a rooted tree* $T$, *the restriction* $i^!_{S_T} \bar{B}^{\mathrm{ch}}_n(\mathcal{A})$ *is acyclic outside degree* $0$.

***Tropical and representation-theoretic:***

(xi) ***Tropical Koszulness.*** *The tropicalization of the bar complex* $\mathrm{trop}(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$, *obtained by passing to the dual complex of the FM compactification and retaining only the combinatorial (tree-level) data, satisfies tropical* $E_2$*-collapse: the tropical spectral sequence has* $d_r^{\mathrm{trop}} = 0$ *for* $r \geq 2$.

(xii) ***Ext diagonal vanishing.*** $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X, \omega_X) = 0$ *for* $p \neq q$, *where the bigrading comes from the PBW filtration on the bar resolution.*

(xiii) ***Kac–Shapovalov determinant nonvanishing.*** *The Shapovalov determinant* $\det G_h \neq 0$ *in the bar-relevant range: the PBW-to-bar comparison map is injective at every bar-relevant conformal weight* $h$.

***Poisson-geometric (affine family):***

(xiv) ***Sklyanin*** $H^2 = 0$***.*** *For* $\mathcal{A} = V_k(\mathfrak{g})$ *with* $\mathfrak{g}$ *semisimple, the second Poisson cohomology of the Sklyanin bracket* $\{-,-\}_{\mathrm{STS}}$ *on* $(\mathfrak{g}^!)^*$ *vanishes:* $H^2_\pi((\mathfrak{g}^!)^*, \{-,-\}_{\mathrm{STS}}) = 0$. *Equivalently, the Sklyanin–Poisson structure determined by the classical* $r$*-matrix* $r(z) = k\Omega/z$ *admits no nontrivial infinitesimal deformations.*

***Conditional and partial:***

(xv) ***Lagrangian criterion*** (conditional on perfectness). $\mathcal{M}_{\mathcal{A}}$ *and* $\mathcal{M}_{\mathcal{A}^!}$ *are transverse Lagrangians in the* $(-1)$*-shifted symplectic deformation space* $\mathcal{M}_{\mathrm{comp}}$.

(xvi) $\mathcal{D}$***-module purity*** (one-directional). *Each* $\bar{B}^{\mathrm{ch}}_n(\mathcal{A})$ *is pure of weight* $n$ *as a mixed Hodge module, with characteristic variety aligned to FM boundary strata. Implies* (x)*. Converse proved for affine Kac–Moody; open in general.*

*Remark* 3.3 (Numbering convention)*.* The numbering (i)–(xiv) in this paper groups conditions by mathematical kinship rather than by the order of the proof. The parent monograph [23] numbers twelve conditions (i)–(xii) in a slightly different order; the present list expands by introducing shadow-tower formality (iv), $E_2$-formality (v), curve independence (vi), PBW universality (vii), and tropical Koszulness (xi) as named conditions, and by separating the Sklyanin criterion (xiv) from the Ext diagonal vanishing (xii). The Lagrangian criterion (xv) and $\mathcal{D}$-module purity (xvi) are kept outside the numbered list because their equivalence class is incomplete.

*Remark* 3.4 (Honest audit of the fourteen). The headline count *fourteen* aggregates conditions of three logical statuses; we enumerate them to avoid the over-claim that all sixteen labeled items (i)–(xvi) are unconditional biconditionals.

**Unconditional biconditionals** (eleven total): (i) Koszul, (ii) PBW $E_2$, (iii) $A_\infty$-formality, (iv) shadow-tower formality, (vi) curve independence, (vii) PBW universality, (viii) Barr–Beck–Lurie monadicity, (ix) factorization-homology concentration, (x) FM boundary acyclicity, (xii) Ext diagonal, (xiii) Kac–Shapovalov determinant.

**One-directional implications** (two total): (v) $E_2$-formality of $\mathrm{ChirHoch}^*$: direction (i)⇒(v) proved, converse closed below in Theorem 3.5. (xvi) $\mathcal{D}$-module purity: (xvi)⇒(x) proved, converse for affine Kac–Moody only.

**Conditional on hypothesis** (two total): (xi) tropical Koszulness, equivalent when the FM dual complex is simplicial (true for smooth curves). (xv) Lagrangian criterion, equivalent under perfectness and non-degeneracy of the ambient tangent complex.

**Family-restricted** (one total): (xiv) Sklyanin $H^2 = 0$, biconditional for the affine Kac–Moody family, one-directional (⇒(ii)) in general.

Count: $11+2+2+1 = 16$ labeled items. The headline *fourteen* corresponds to (i)–(xiv); the conditional pair (xv), (xvi) is kept outside. After Theorem 3.5 the unconditional count promotes to twelve.

**Theorem 3.5** (Converse of (v): $E_2$-formality of $\mathrm{ChirHoch}^*$ implies chiral Koszulness)**.** *Let $\mathcal{A}$ be a positive-energy chiral algebra on $X$ with PBW filtration. Suppose condition (v) holds in its full form:* $\mathrm{ChirHoch}^*(\mathcal{A})$ *is concentrated in cohomological degrees $\{0, 1, 2\}$, carries the polynomial Hilbert series $P_{\mathcal{A}}(t) = \dim Z(\mathcal{A}) + \dim \mathrm{ChirHoch}^1(\mathcal{A}) \cdot t + \dim Z(\mathcal{A}^!) \cdot t^2$, and is formal as an $E_2$-algebra. Then $\mathcal{A}$ is chirally Koszul (condition (i)). Consequently, (v) is an unconditional biconditional, upgrading the unconditional count from eleven to twelve.*

*Proof.* Four steps.

*Step 1: Chiral Deligne-Tamarkin identification.* Under Theorem H in the chiral-higher-Deligne form of Vol II [**?**, §chiral-higher-deligne], the $E_2$-algebra structure on $\mathrm{ChirHoch}^*(\mathcal{A})$ is identified with the derived $E_2$-chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch},E_2}(\mathcal{A})$ via

$$\mathrm{ChirHoch}^*(\mathcal{A}) \;\simeq\; R\mathrm{Hom}_{E_2^{\mathrm{ch}}\text{-mod}}(\mathcal{A}, \mathcal{A}),$$

where the right-hand side is computed in the factorization ambient on $\mathbb{P}^1$. This identification is associator-dependent; fix a Drinfeld associator $\Phi$ (e.g. $\Phi_{\mathrm{KZ}}$).

*Step 2: Cobar of formal $E_2$-algebra.* The $E_2$-formality hypothesis on $\mathrm{ChirHoch}^*(\mathcal{A})$ gives a strict $E_2$-algebra model. Kontsevich–Tamarkin operadic Koszul duality between $E_2$ and its shifted dual $E_2\{1\}$ produces an $E_2$-cobar $\Omega_{E_2}(\mathrm{ChirHoch}^*(\mathcal{A}))$ that computes $R\mathrm{Hom}_{E_2\text{-mod}}(k, \mathcal{A})$ in the chiral ambient. The polynomial Hilbert series hypothesis ($\dim = \dim Z(\mathcal{A}) +$

$\dim \mathrm{ChirHoch}^1 \cdot t + \dim Z(\mathcal{A}^!) \cdot t^2$, no higher) forces the brace operations $\mathrm{br}_n$ on $\mathrm{ChirHoch}^*(\mathcal{A})$ to vanish for all $n \geq 3$: any nonzero $\mathrm{br}_n$ would contribute a class in degree $\geq 3$ by arity-counting on $\overline{C}_n(\mathbb{C})$-chains, contradicting concentration in $\{0, 1, 2\}$.

*Step 3: Transfer to $A_\infty$-operations on bar cohomology.* The bar coalgebra $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ and its dual $R\mathrm{Hom}(\bar{B}^{\mathrm{ch}}(\mathcal{A}), k) = C^*_{\mathrm{ch}}(\mathcal{A}, k)$ are related to $\mathrm{ChirHoch}^*(\mathcal{A})$ by the chiral cap product. Under the chiral Deligne identification of Step 1, the transferred $A_\infty$-operations $m_n$ on $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ are images of the brace operations $\mathrm{br}_n$ under the Koszul dual pairing. Brace vanishing for $n \geq 3$ (Step 2) forces $m_n = 0$ for all $n \geq 3$. This is condition (iii), $A_\infty$-formality of bar cohomology.

*Step 4: Close the circuit.* By the core circuit of Theorem 3.2, (iii) $\Leftrightarrow$ (i): $A_\infty$-formality of bar cohomology is equivalent to chiral Koszulness. Hence (v)$\Rightarrow$(iii)$\Rightarrow$(i), closing the converse.

The proof requires the associator choice $\Phi$ of Step 1, so the converse is proved on the full $\mathrm{GRT}_1$-torsor $M_{\mathrm{Kosz}}(\mathcal{A})$: once any Koszulness characterization holds at one chart, it holds at every chart. On the $\Phi_{\mathrm{KZ}}$-chart the brace-vanishing computation of Step 2 is immediate; Tamarkin $\Phi$-transfer propagates to other charts. □

*Remark* 3.6 (Dependence on chiral Higher Deligne). Theorem 3.5 uses the chiral Higher Deligne identification $\mathrm{ChirHoch}^*(\mathcal{A}) \simeq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch},E_2}(\mathcal{A})$ established in the Vol II reconstitution (`chapters/theory/ chiral_higher_deligne.tex`, `thm:chd-deligne-tamarkin`). The associator-independent formulation remains open; the associator-dependent form used here is sufficient to close the converse on the full $\mathrm{GRT}_1$-torsor.

*Remark* 3.7 (The four objects). The four objects of chiral Koszul duality must not be conflated:

(1) $\bar{B}^{\mathrm{ch}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$: the bar *coalgebra*, with deconcatenation coproduct and bar differential.
(2) $\mathcal{A}^i = H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$: the dual *coalgebra*, Koszul cohomology of the bar complex.
(3) $\mathcal{A}^! = ((\mathcal{A}^i)^\vee)$: the Koszul dual *algebra*, obtained by Verdier duality.
(4) $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = R\mathrm{Hom}_{\mathcal{A}\otimes\mathcal{A}^{\mathrm{op}}}(\mathcal{A}, \mathcal{A})$: the derived chiral center, computed using the bar complex as a resolution.

The bar complex is the $E_1$ engine; $\Omega(\bar{B}^{\mathrm{ch}}(\mathcal{A})) = \mathcal{A}$ is *inversion*, not Koszul duality. The Koszul dual comes from Verdier duality. The derived center comes from chiral Hochschild cochains.

## 4. The circuit of implications

The proof of Theorem 3.2 is organized as a directed graph of implications. The central structure is the *core circuit*, a cycle of biconditional implications among five conditions:

$$\text{(i)} \;\Leftrightarrow\; \text{(ii)} \;\Leftrightarrow\; \text{(iii)} \;\Leftrightarrow\; \text{(viii)} \;\Leftrightarrow\; \text{(ix)}. \tag{4.1}$$

Each of the remaining conditions is shown equivalent to a designated node of this core by a satellite implication. The full graph has the following shape.

$$
\begin{tikzcd}
 & \text{(iv) shadow form.} \arrow[d, Leftrightarrow] & & \\
\text{(xii) Ext diag.} \arrow[r, Leftrightarrow] & \text{(ii) PBW } E_2 \arrow[r, Leftrightarrow] \arrow[dl, Leftrightarrow] \arrow[ddl, Leftrightarrow] \arrow[dddl, Leftrightarrow] \arrow[dd, Leftrightarrow] \arrow[dddd, Leftrightarrow, bend left] & \text{(i) Koszul} \arrow[r, Leftrightarrow] & \text{(viii) BBL} \\
\text{(xiii) Kac–Shap.} & \text{(iii) } A_\infty \text{ form.} & & \\
\text{(vi) curve indep.} & \text{(vii) PBW univ.} \arrow[ddd, dashed] & & \\
\text{(x) FM bdry} \arrow[r, Leftrightarrow] & \text{(xi) tropical} & & \\
\text{(v) } E_2\text{-Hoch} & \text{(ix) FH conc.} & & \\
\text{(xiv) Skl. } H^2 \arrow[uuuu, Rightarrow] & \text{(xv) Lagr.} & \text{(xvi) purity} \arrow[uuuul, dashed] &
\end{tikzcd}
$$

Dashed arrows indicate conditional or partial implications. We now give proof sketches for each link, following the order of the graph from the core outward.

### 4.1. **The core circuit.**

(i) $\Leftrightarrow$ (ii): *Koszulness and PBW collapse.* Koszulness is acyclicity of $K_\tau^L(\mathcal{A}, \bar{B}^{\mathrm{ch}}(\mathcal{A}))$ for the canonical twisting morphism $\tau_0$. The left twisted tensor product is filtered by the PBW bar degree, and on the associated graded the twisted complex becomes the standard Koszul complex of the Poisson vertex algebra $R_{\mathcal{A}}$. Acyclicity on the associated graded lifts to acyclicity of $K_\tau^L$ if and only if the spectral sequence collapses at $E_2$ [23, Thm. 4.2.1]. This is the PBW Koszulness criterion.

Conversely, $E_2$-collapse means $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ is concentrated on the PBW diagonal $p = q$. A $d_r$-differential in the spectral sequence moves off-diagonal by $(r, 1-r)$; off-diagonal classes in the abutment would contradict diagonal concentration. The spectral sequence converges conditionally in the sense of Boardman (the PBW filtration on a positive-energy chiral algebra is bounded below and exhaustive), and in characteristic zero with a split filtration the extension problem is trivial.

(ii) $\Leftrightarrow$ (iii): *PBW collapse and $A_\infty$-formality.* The Homological Perturbation Lemma identifies the differentials $d_r$ of the PBW spectral sequence with the transferred $A_\infty$-operations $m_{r+1}$ on bar cohomology, up to sign. Hence

$d_r = 0$ for all $r \geq 2$ is equivalent to $m_n = 0$ for all $n \geq 3$: the minimal $A_\infty$-model is formal.

For the converse, if $m_n = 0$ for $n \geq 3$, the minimal model is a strict dg algebra. The chiral $A_\infty$-structure is computed fiberwise on each FM stratum; on each fiber, Keller's classicality theorem for formal $A_\infty$-algebras gives $E_2$-collapse. The PBW filtration is defined fiberwise and compatible with the FM stratification, so fiberwise collapse assembles to global collapse.

(i) $\Leftrightarrow$ (viii): *Koszulness and Barr–Beck–Lurie monadicity.* The Barr–Beck–Lurie theorem [19, Thm. 4.7.3.5] states that the comparison functor $\Phi$ for the bar–cobar adjunction is an equivalence if and only if the bar functor is conservative and preserves totalizations of bar-split cosimplicial objects.

On the Koszul locus, the bar functor detects quasi-isomorphisms (the bar–cobar counit inverts them), giving conservativity. The bar–cobar adjunction is a Quillen equivalence on the Koszul locus [23, §5]; a Quillen equivalence of stable model categories satisfies the monadic descent conditions of [19, Prop. 4.7.3.16]. Hence BBL monadicity holds if and only if the chiral algebra is Koszul.

(i) $\Leftrightarrow$ (ix): *Koszulness and factorization homology concentration.* The bar complex computes factorization homology: $\bar{B}^{\mathrm{ch}}_n(\mathcal{A}) = (\int_X \mathcal{A})_n$. At genus zero, $E_2$-collapse gives $H^k(\int_{\mathbb{P}^1} \mathcal{A}) = 0$ for $k \neq 0$: the Goodwillie filtration has associated graded given by bar contributions, each concentrated in degree 0 by PBW collapse. On the uniform-weight lane, the same holds at every genus: the loop-order spectral sequence collapses on the modular Koszul locus, each vertex contribution remains in degree 0, and the surviving scalar is $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ (UNIFORM-WEIGHT).

Conversely, concentration at genus zero specializes to $H^k(\int_{\mathbb{P}^1} \mathcal{A}) = 0$ for $k \neq 0$. This is computed by the bar complex: the geometric realization $\int_{\mathbb{P}^1} \mathcal{A} \simeq |\bar{B}^{\mathrm{ch}}(\mathcal{A})|$. Concentration in degree 0 forces the bar spectral sequence to collapse, which by the first equivalence is Koszulness.

## 4.2. **Satellite equivalences from the core.**

(ii) $\Leftrightarrow$ (iv): *PBW collapse and shadow-tower formality.* The shadow obstruction tower $\{o_{r+1}(\mathcal{A})\}_{r\geq 2}$ has two projections: the genus-zero slice $o^{(0)}_{r+1}(\mathcal{A})$ and the all-genera projection. Shadow-tower formality (vanishing of the genus-zero slice) is identified with the transferred $L_\infty$-brackets $\ell^{(0)}_{r+1}$ of the modular convolution algebra at genus zero. The HPL transfer identifies these with the PBW spectral sequence differentials. Hence shadow-tower formality is equivalent to $E_2$-collapse.

The all-genera shadow tower, by contrast, can be nontrivial even for Koszul algebras: this is the shadow depth classification G/L/C/M. Shadow-tower formality is the genus-zero condition; shadow depth is the all-genera condition. They are independent invariants.

(i) $\Rightarrow$ (v): *Koszulness implies $E_2$-formality of ChirHoch.* The bar–cobar quasi-isomorphism $\Omega^{\mathrm{ch}}(\bar{B}^{\mathrm{ch}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ provides a free resolution of $\mathcal{A}$. Computing $\mathrm{ChirHoch}^*(\mathcal{A})$ through this resolution, the bar–cobar filtration forces diagonal concentration. The polynomial structure in degrees $\{0, 1, 2\}$ follows: degree 0 is the center, degree 1 is first-order deformations, degree 2 is obstructions. The $E_2$-formality of $\mathrm{ChirHoch}^*(\mathcal{A})$ uses the formality of the local configuration spaces $\overline{C}_n(\mathbb{C})$.

This is a one-way consequence: $E_2$-formality of $\mathrm{ChirHoch}^*$ does not force the underlying graded-commutative algebra to be free, so the converse from (v) to bar–cobar inversion is not established.

(ii) $\Leftrightarrow$ (vi): *PBW collapse and curve independence.* Curve independence at genus zero follows from PBW collapse because the collapsed spectral sequence depends only on the associated graded $R_{\mathcal{A}}$, which is a Poisson vertex algebra independent of the global geometry of $X$. The FM compactification $\overline{C}_n(X)$ depends on $X$, but the $E_2$-page $H^{p+q}(\mathrm{gr}_p^F \bar{B}^{\mathrm{ch}}(\mathcal{A}))$ is computed from $R_{\mathcal{A}}$ by Chevalley–Eilenberg methods that see only the local OPE data.

Conversely, if factorization homology is curve-independent at genus zero, the bar complex cannot have differentials sensitive to global geometry. The PBW spectral sequence differentials $d_r$ for $r \geq 2$ involve integration over FM strata of codimension $r$; these strata encode collisions of $r + 1$ points and are sensitive to the curve's geometry at the $r$-fold collision scale. Curve independence forces these to vanish, giving $E_2$-collapse.

(iii) $\Leftrightarrow$ (vii): *$A_\infty$-formality and PBW universality.* PBW universality states that the PBW filtration deforms flatly with classically Koszul associated graded. If $\mathcal{A}$ is freely strongly generated (the generators form a free $\mathcal{D}_X$-module), the associated graded is a free polynomial PVA, which is quadratic-Koszul. The flat deformation property then implies that the bar spectral sequence for $\mathcal{A}_\hbar$ is a deformation of the (collapsed) spectral sequence for $\mathcal{A}_0$. By semicontinuity, the collapsed page persists, giving $E_2$-collapse and hence $A_\infty$-formality.

Conversely, $A_\infty$-formality of bar cohomology means the transferred operations $m_n = 0$ for $n \geq 3$; this is the linearization condition that PBW universality encodes.

(i) $\Leftrightarrow$ (x): *Koszulness and FM boundary acyclicity.* Each FM boundary stratum $S_T$ is indexed by a rooted tree $T$. The residue component $d_{\mathrm{res}}|_{S_T}$ of the bar differential restricts to the tensor product of lower-degree bar complexes at the vertices of $T$. $E_2$-collapse at each vertex gives acyclicity at $S_T$: $H^k(i^!_{S_T} \bar{B}^{\mathrm{ch}}_n(\mathcal{A})) = 0$ for $k \neq 0$.

Conversely, acyclicity at the deepest strata (binary collision divisors $S = D_{\{i,j\}} \cong X \times \overline{C}_{n-1}(X)$) forces every OPE residue to be exact in the PBW-graded sense, which is the condition for $E_2$-collapse. Iterating over all strata closes the equivalence.

(x) $\Leftrightarrow$ (xi): *FM boundary acyclicity and tropical Koszulness.* The tropicalization of the bar complex retains only the combinatorial data visible in the dual complex of the FM compactification: the rooted trees indexing boundary strata, the face maps between strata, and the residues of the bar differential along each stratum. Tropical $E_2$-collapse is the assertion that these combinatorial residues produce no off-diagonal cohomology.

FM boundary acyclicity (the analytic statement) implies tropical Koszulness (the combinatorial statement) by restriction. The converse holds when the dual complex is simplicial (the standard case for smooth curves): the combinatorial data determines the analytic behavior up to homotopy equivalence, so tropical acyclicity forces analytic acyclicity.

(i) $\Leftrightarrow$ (xii): *Koszulness and Ext diagonal vanishing.* $E_2$-collapse places $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ on the PBW diagonal $p = q$. The bar complex computes $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X, \omega_X)$, so diagonal concentration of bar cohomology is Ext diagonal vanishing.

Conversely, Ext diagonal vanishing means the abutment of the PBW spectral sequence is diagonal. Any nonzero differential $d_r$ for $r \geq 2$ would produce off-diagonal classes, contradicting diagonal concentration. Hence $d_r = 0$ for all $r \geq 2$.

(i) $\Leftrightarrow$ (xiii): *Koszulness and Kac–Shapovalov determinant.* If $\mathcal{A}$ is Koszul, the PBW filtration is strict on the vacuum module: $\mathrm{gr}^F M_0(\mathcal{A}) \cong \mathrm{Sym}^{\mathrm{ch}}(V)$. Strictness implies non-degeneracy of the Shapovalov form $G_h$ in the bar-relevant range.

Conversely, non-degeneracy of $G_h$ makes the PBW-to-bar comparison injective at every bar-relevant weight. Suppose $d_r \neq 0$ for some $r \geq 2$; then there exists a nonzero class in $E_r^{p,q}$ with nonzero differential, whose lift to the bar complex contradicts the acyclicity of $\mathrm{Sym}(V)$ as a Koszul complex. Hence $d_r = 0$ for all $r \geq 2$.

(xiv) $\Rightarrow$ (ii): *Sklyanin $H^2 = 0$ implies PBW collapse (affine family).* For $\mathcal{A} = V_k(\mathfrak{g})$ with $\mathfrak{g}$ semisimple, the Koszul dual coalgebra $\bar{B}^{\mathrm{ch}}(\mathcal{A})^!$ carries the Poisson-Lie group $(\mathfrak{g}^!)^*$ with the Sklyanin–Poisson bracket determined by the classical $r$-matrix $r(z) = k\Omega/z$ via the Semenov-Tian-Shansky construction. The second Poisson cohomology $H^2_\pi$ controls infinitesimal deformations of this bracket.

The vanishing $H^2_\pi = 0$ is established by Whitehead's second lemma (for $\mathfrak{g}$ semisimple, $H^2(\mathfrak{g}; M) = 0$ for finite-dimensional modules $M$): the Poisson cochain complex $(\mathfrak{X}^\bullet, \delta_\pi)$ is the associated graded of the bar differential under PBW filtration, so vanishing of $H^2_\pi$ is the associated-graded avatar of $E_2$-collapse. The proof: five independent derivation paths (direct kernel/image, Chevalley–Eilenberg comparison, polynomial-degree spectral sequence, Whitehead reduction, and numerical from the compute engine) confirm $H^0_\pi = \mathbb{C}[C_2]$ (Casimir ring), $H^1_\pi = H^2_\pi = H^3_\pi = 0$.

This is proved as a biconditional for the affine Kac–Moody family. For general chiral algebras, the Sklyanin construction does not apply (it requires a Lie-algebraic $r$-matrix), so the equivalence is restricted to this family.

(xv): *Lagrangian criterion, conditionally.* Under the perfectness and non-degeneracy hypotheses on the ambient tangent complex: on the Koszul locus, the complementarity splitting

$$\mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^!) \;\simeq\; H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$$

identifies $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ as complementary subspaces of $\mathcal{M}_{\mathrm{comp}}$. The $(-1)$-shifted symplectic structure makes each isotropic; complementarity gives maximal dimension, hence Lagrangian. Conversely, transverse Lagrangian intersection in a $(-1)$-shifted symplectic space has expected dimension zero; its discreteness is acyclicity of the twisted tensor product, which is Koszulness.

(xvi) $\Rightarrow$ (x): *$\mathcal{D}$-module purity implies FM boundary acyclicity.* Suppose each $\bar{B}^{\mathrm{ch}}_n(\mathcal{A})$ is pure of weight $n$ as a mixed Hodge module. By Saito's theory, the weight filtration on a pure Hodge module splits up to semisimplification; purity plus the alignment of the characteristic variety to FM strata forces acyclicity outside degree 0 on every stratum. The converse (Koszulness implies purity) is proved for the affine Kac–Moody family, where the bar complex is identified with a geometric complex whose purity follows from the Riemann hypothesis for weight filtrations.

*Remark* 4.1 (Proof web redundancy)*.* The proof graph is intentionally overdetermined. The core circuit has five nodes and can be verified by tracing any four of the five biconditionals. Each satellite equivalence is independent: a failure in one link does not compromise the others. The total number of proved implications (counting both directions of each biconditional) is twenty-four unconditional and four conditional.

## 5. Scalar saturation

5.1. **The modular characteristic.** Let $\mathcal{A}$ be a chiral algebra on $X$. The universal Maurer–Cartan element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ of the bar differential lives in the modular convolution dg Lie algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ and satisfies

$$d\,\Theta_{\mathcal{A}} + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0. \tag{5.1}$$

At each degree $r \geq 2$, $\Theta_{\mathcal{A}}$ projects to a shadow invariant $S_r(\mathcal{A})$.

The $E_1$ ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is the primitive object: it carries the full $R$-matrix $r_{\mathcal{A}}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$. The symmetric bar $\bar{B}^{\Sigma}(\mathcal{A})$ is the $\Sigma_n$-coinvariant quotient: a lossy projection that discards the matrix-valued data and retains only the scalar

$$\kappa(\mathcal{A}) \;=\; S_2(\mathcal{A}), \tag{5.2}$$

the *modular characteristic*: the single number that survives averaging.

*Remark* 5.1 (Family-specific formulas). The modular characteristic is family-specific; writing it from memory is forbidden. The canonical formulas, verified by multiple independent paths, are:

| Family | $\kappa(\mathcal{A})$ | Sanity checks |
|---|---|---|
| Heisenberg $\mathcal{H}_k$ | $k$ | $k=0 \to 0$; $k=1 \to 1$ |
| Affine KM $V_k(\mathfrak{g})$ | $\dfrac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee}$ | $k=0 \to \frac{\dim(\mathfrak{g})}{2}$; $k=-h^\vee \to 0$ |
| Virasoro $\mathrm{Vir}_c$ | $c/2$ | $c=0 \to 0$; $c=13 \to 13/2$ |
| $\mathcal{W}_N$ (principal) | $c \cdot (H_N - 1)$ | $N=2 \to c/2 = \kappa^{\mathrm{Vir}}$; $N=3 \to 5c/6$ |

where $H_N = \sum_{j=1}^N 1/j$ is the $N$-th harmonic number.

5.2. **The saturation theorem.**

**Theorem 5.2** (Scalar saturation)**.** *Let $\mathcal{A}_k$ be a one-parameter algebraic family of chiral algebras with rational OPE coefficients, parametrized by a level $k$. At degree 2, the universal MC element is determined by the single scalar $\kappa(\mathcal{A}_k)$:*

$$\Theta^{(2)}_{\mathcal{A}_k} \;=\; \kappa(\mathcal{A}_k) \cdot \eta \otimes \Lambda, \tag{5.3}$$

*where $\eta = d\log(z_1 - z_2)$ is the Arnold propagator and $\Lambda$ is the universal quadratic Casimir element.*

*Proof sketch.* The degree-two component of $\Theta_{\mathcal{A}}$ in the modular convolution algebra takes values in $H^2(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}; \mathrm{Sym}^q V)$ for appropriate $q$. Whitehead's second lemma gives vanishing of this cohomology for reductive Lie algebras at generic level: the space of degree-two MC deformations is one-dimensional, spanned by $\eta \otimes \Lambda$. The coefficient is determined by the $\Sigma_n$-coinvariant projection: $\mathrm{av}(\Theta^{(2)}_{\mathcal{A}}) = \kappa(\mathcal{A}) \cdot [\lambda_1]$, which fixes the coefficient as $\kappa$.

The argument has three steps:

(1) **Whitehead reduction.** For each standard family, the Lie algebra cohomology $H^2(\mathfrak{g}; \mathrm{Sym}^q V) = 0$ at generic level (the reductive hypothesis is used here). This gives a one-dimensional deformation space at degree two.
(2) **Algebraic semicontinuity.** The exceptional locus where Whitehead reduction fails is Zariski-closed in the level parameter $k$. Since the free-field limit $k \to 0$ (for families where $\kappa(k=0) = 0$) provides a base point where saturation holds trivially, the exceptional locus is empty.
(3) **Coefficient identification.** The averaging map $\mathrm{av}\colon \mathfrak{g}^{E_1}_{\mathcal{A}} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ projects $\Theta^{(2)}_{\mathcal{A}}$ to $\kappa(\mathcal{A}) \cdot [\lambda_1]$. Since av is the $\Sigma_n$-coinvariant projection and $\eta \otimes \Lambda$ is the unique $\Sigma_2$-invariant degree-two element (up to scalar), the coefficient is $\kappa$.

For non-abelian affine KM, the averaging identity involves the Sugawara shift: $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2 = \kappa(V_k(\mathfrak{g}))$ where $\mathrm{av}(r(z)) = k \cdot \dim(\mathfrak{g})/(2h^\vee)$

is the double-pole channel and $\dim(\mathfrak{g})/2$ is the simple-pole self-contraction through the adjoint Casimir. □

*Remark* 5.3 (Scope). Scalar saturation at degree 2 holds for every family in the standard landscape at all non-critical levels, including admissible. For multi-weight algebras at genus $g \geq 2$, saturation holds at degree 2 but the full MC element receives cross-channel corrections: the stronger identity $\Theta_{\mathcal{A}}^{\min} = \kappa \cdot \eta \otimes \Lambda$ is proved on the uniform-weight lane and fails at $g \geq 2$ for multi-weight families (ALL-WEIGHT $+\delta F_g^{\text{cross}}$).

5.3. **From scalar saturation to Koszulness.** Scalar saturation establishes that $\kappa$ controls the MC element at degree 2. The question is what controls it at *all* degrees. The answer is the Koszul condition.

**Proposition 5.4** (Koszulness controls the full tower)**.** *If $\mathcal{A}$ is chirally Koszul, then the universal MC element $\Theta_{\mathcal{A}}$ is determined at every degree by its genus-zero projection. Concretely:*

1. *The shadow invariants $S_r(\mathcal{A})$ for $r \geq 3$ are determined by the $A_\infty$-formality data of bar cohomology.*
2. *On the uniform-weight lane, $\Theta_{\mathcal{A}}$ is determined by $\kappa(\mathcal{A})$ and the shadow depth class.*
3. *For class G algebras (Heisenberg), $S_r = 0$ for $r \geq 3$: the MC element is entirely controlled by $\kappa$.*
4. *For class M algebras (Virasoro), $S_r \neq 0$ for infinitely many $r$, but each $S_r$ is determined by $\kappa = c/2$ and $S_4 = 10/[c(5c + 22)]$.*

*Proof sketch.* $A_\infty$-formality (condition (iii) of Theorem 3.2) means $m_n = 0$ for $n \geq 3$: the bar cohomology algebra has no higher operations. The MC moduli of a formal $A_\infty$-algebra is discrete; the shadow invariants $S_r$ are the Taylor coefficients of the unique MC solution in the completion of the convolution algebra. Each $S_r$ is determined recursively from $\kappa = S_2$ and the structure constants of the algebra.

For class G, all $S_r = 0$ for $r \geq 3$ by direct computation (the bar complex is abelian). For class M, the recursion is controlled by the Riccati equation $\Delta = 8\kappa S_4$, where $\Delta = 0$ characterizes class G. The seed $S_4$ determines $S_r$ for all $r$ via the shadow algebra Lie bracket; for Virasoro, $S_4 = 10/[c(5c+22)] \sim 2/c^2$ at large $c$. □

*Remark* 5.5 (The critical discriminant). The discriminant $\Delta = 8\kappa \cdot S_4$ is linear in $\kappa$ (not quadratic). $\Delta = 0$ if and only if the shadow tower is finite. For $\kappa \neq 0$, finiteness is equivalent to $S_4 = 0$, which characterizes class G. For $\kappa = 0$ (critical level), $\Delta = 0$ regardless of $S_4$.

5.4. **The $E_1$ structure.** Scalar saturation is a statement about the $\Sigma_n$-coinvariant quotient. The full $E_1$ ordered bar complex carries strictly more data: the matrix-valued classical $r$-matrix $r_{\mathcal{A}}(z) \in \operatorname{End}_{\mathcal{A}}(2)[\![z^{-1}]\!]$, of which $\kappa$ is only the scalar shadow $\kappa = \operatorname{av}(r_{\mathcal{A}}(z))$.

The $r$-matrix satisfies the classical Yang–Baxter equation $[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0$ and determines the Yangian structure on the ordered bar cohomology. It absorbs one pole from the OPE via the logarithmic propagator: for Heisenberg, the OPE has a double pole and the $r$-matrix has a simple pole, $r^{\mathrm{Heis}}(z) = k/z$. For Virasoro, the OPE has a quartic pole and the $r$-matrix has a cubic pole, $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$. For affine KM in the trace-form convention, $r^{\mathrm{KM}}(z) = k\Omega/z$, where $\Omega$ is the inverse Killing form Casimir and the level prefix $k$ is mandatory: at $k = 0$ the $r$-matrix vanishes (abelian limit).

The averaging map $\mathrm{av}\colon \mathfrak{g}_{\mathcal{A}}^{E_1} \to \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is the $\Sigma_n$-coinvariant projection. It is lossy: the full $r$-matrix determines $\kappa$, but $\kappa$ does not determine the $r$-matrix. The five theorems of the parent monograph extract the $\Sigma_n$-invariant content; the present paper works at this invariant level, where scalar saturation gives full control at degree two and Koszulness extends control to all degrees.

## 6. The standard landscape

The fourteen characterizations of Theorem 3.2 are verified on every family in the standard landscape. The verification proceeds family by family, each through a different Koszul equivalence: the structure of the algebra determines which criterion provides the cleanest proof.

**Theorem 6.1** (Every standard family is chirally Koszul)**.** *The following chiral algebras are chirally Koszul: Heisenberg $\mathcal{H}_k$ at all levels $k$; lattice algebras $V_L$ for all even lattices $L$; affine Kac–Moody $V_k(\mathfrak{g})$ for all semisimple $\mathfrak{g}$ at all non-critical levels $k \neq -h^\vee$; Virasoro $\mathrm{Vir}_c$ at all $c \neq 0$; the principal $\mathcal{W}$-algebra $\mathcal{W}_N$ at all $N$ and generic level; the $\beta\gamma$ system at standard conformal weight; and the Bershadsky–Polyakov algebra $\mathcal{W}_k^{(2)}(\mathfrak{sl}_3)$ at generic level.*

The proof is the content of this section. Table 1 records the family, the Koszul equivalence that proves it, the shadow class, the modular characteristic, and the Koszul complementarity conductor.

### 6.1. Heisenberg.

The Heisenberg algebra $\mathcal{H}_k$ has a single generator $J$ with OPE $J(z)J(w) \sim k/(z-w)^2$, so the bar complex $\bar{B}^{\mathrm{ch}}(\mathcal{H}_k)$ is the desuspended tensor coalgebra on a one-dimensional augmentation ideal. The PBW spectral sequence has a single nonzero column at $p = 1$, so $E_2$-collapse is vacuous: there are no possible higher differentials. This is equivalence (ii) of Theorem 3.2, in its trivial form. The modular characteristic is $\kappa(\mathcal{H}_k) = k$, and the shadow tower terminates at $r = 2$: class G. The Koszul dual is $\mathcal{H}_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$ (the chiral symmetric algebra on the linear dual), with $\kappa(\mathcal{H}_k^!) = -k$, giving $K = k + (-k) = 0$.

### 6.2. Lattice algebras.

The lattice vertex algebra $V_L$ associated to an even lattice $L$ of rank $r$ is an extension of the Heisenberg algebra by vertex operators $e^\alpha$ for $\alpha \in L$. The PBW filtration on $V_L$ has associated graded $\mathrm{gr}^F V_L \cong \mathrm{Sym}^{\mathrm{ch}}(\mathfrak{h}) \otimes \mathbb{C}[L]$, where $\mathfrak{h} = L \otimes_{\mathbb{Z}} \mathbb{C}$. The lattice grading gives a

| Family | Koszul equivalence | Class | $\kappa$ | $K$ |
|---|---|---|---|---|
| $\mathcal{H}_k$ | (ii) PBW, trivial | G | $k$ | 0 |
| Lattice $V_L$ | (ii) PBW, lattice | G | $\mathrm{rk}(L)/2$ | 0 |
| $V_k(\mathfrak{g})$ | (xiii) Kac–Shapovalov | L | $\frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee}$ | 0 |
| $\mathrm{Vir}_c$ | (iv) shadow formality | M | $c/2$ | 13 |
| $\mathcal{W}_N$ | (xiii) via DS from KM | M | $c(H_N-1)$ | family-dep. |
| $\beta\gamma$ | (iii) $A_\infty$-formality | C | $6\lambda^2-6\lambda+1$ | 0 |
| $\mathcal{W}_k^{(2)}(\mathfrak{sl}_3)$ | (xii) Ext diag. | M | $c(k)/6$ | 98/3 |

Table 1. The standard landscape and Koszulness. $H_N = \sum_{j=1}^N 1/j$. The column $K$ denotes the Koszul complementarity conductor $K = \kappa(\mathcal{A}) + \kappa(\mathcal{A}^!)$.

decomposition of the bar complex into weight spaces, each of which reduces to a Heisenberg bar complex tensored with the group algebra. The PBW spectral sequence collapses at $E_2$ in each weight sector: equivalence (ii). The modular characteristic is $\kappa(V_L) = \mathrm{rk}(L)/2$, and the shadow class is G.

6.3. **Affine Kac–Moody.** For $\mathcal{A} = V_k(\mathfrak{g})$ with $\mathfrak{g}$ semisimple and $k \neq -h^\vee$, the Koszulness proof goes through the Kac–Shapovalov criterion (xiii). The Shapovalov form $G_h$ on the vacuum Verma module $M_0(V_k(\mathfrak{g}))$ is nondegenerate for $k$ outside the critical hyperplane: the Shapovalov determinant is a product over positive roots and PBW-relevant weights, and its zeros are confined to the critical level and the reducibility locus [20, Thm. 11.4].

At non-critical level, $\det G_h \neq 0$ in the bar-relevant range: the PBW-to-bar comparison is injective, and the PBW spectral sequence collapses at $E_2$ by Theorem 3.2 (xiii)$\Leftrightarrow$(ii). This is the chiral analogue of the Garland–Lepowsky concentration theorem [17]: the cohomology of the bar complex is concentrated on the PBW diagonal, the higher differentials vanish, and the associated graded computes the Koszul dual coalgebra.

The modular characteristic is $\kappa(V_k(\mathfrak{g})) = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$. Sanity checks: $k = 0$ gives $\kappa = \dim(\mathfrak{g})/2$ (the abelian limit, not zero: the double-pole vanishes but the simple-pole self-contraction persists); $k = -h^\vee$ gives $\kappa = 0$ (critical level, where Koszulness fails). The shadow class is L: the cubic shadow $S_3 \neq 0$ detects the Lie bracket, but $S_4 = 0$, so the tower terminates at degree 3. The Koszul conductor is $K = \kappa(V_k(\mathfrak{g})) + \kappa(V_{k'}(\mathfrak{g})) = 0$ where $k' = -2h^\vee - k$.

6.4. **Virasoro.** The Virasoro algebra $\mathrm{Vir}_c$ has OPE pole order 4 and is not quadratic in any classical sense. Its Koszulness is proved through shadow-tower formality (iv). The PBW spectral sequence for $\mathrm{Vir}_c$ has higher differentials that could a priori be nonzero; the proof that they vanish uses the recursive structure of the Maurer–Cartan element.

At genus zero, the shadow obstruction tower has $o^{(0)}_{r+1} = 0$ for all $r \geq 2$: the transferred $L_\infty$-brackets $\ell^{(0)}_{r+1}$ vanish because the Virasoro vacuum module satisfies Kac determinant nonvanishing at generic $c$. Shadow-tower formality at genus zero is equivalence (iv)$\Leftrightarrow$(ii): the PBW spectral sequence collapses.

The modular characteristic is $\kappa(\mathrm{Vir}_c) = c/2$, and the shadow class is M: $S_4 = 10/[c(5c + 22)] \neq 0$ for generic $c$, so the all-genera tower is infinite. The Koszul dual is $\mathrm{Vir}^!_c = \mathrm{Vir}_{26-c}$, giving $K = c/2 + (26 - c)/2 = 13$. The self-dual point is $c = 13$ (not $c = 26$).

6.5. **Principal $\mathcal{W}$-algebras.** The principal $\mathcal{W}$-algebra $\mathcal{W}_N$ at level $k$ is obtained from $V_k(\mathfrak{sl}_N)$ by Drinfeld–Sokolov reduction with respect to the principal nilpotent $f_{\mathrm{prin}}$. The Koszulness of $\mathcal{W}_N$ is derived from the Koszulness of the parent affine KM algebra through the functoriality of DS reduction with respect to the bar complex.

DS reduction preserves the Shapovalov criterion: it sends the Kac–Shapovalov form on $V_k(\mathfrak{sl}_N)$ to the $\mathcal{W}$-algebra Shapovalov form on $\mathcal{W}_N$, and nonvanishing of the former at generic level implies nonvanishing of the latter. This gives Koszulness of $\mathcal{W}_N$ through equivalence (xiii). At $N = 2$, this recovers the Virasoro result through a different path (DS from $\mathfrak{sl}_2$ rather than shadow-tower formality).

The generators of $\mathcal{W}_N$ have conformal weights $\{2, 3, \ldots, N\}$: that is $N - 1$ generators, with highest weight $N$. The modular characteristic is $\kappa(\mathcal{W}_N) = c \cdot (H_N - 1)$ where $H_N = \sum_{j=1}^{N} 1/j$. Sanity check at $N = 2$: $H_2 - 1 = 3/2 - 1 = 1/2$, so $\kappa(\mathcal{W}_2) = c/2$, matching $\kappa(\mathrm{Vir}_c)$. At $N = 3$: $H_3 - 1 = 11/6 - 1 = 5/6$, so $\kappa(\mathcal{W}_3) = 5c/6$.

6.6. **The $\beta\gamma$ system.** The $\beta\gamma$ system at conformal weight $\lambda$ has central charge $c_{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1)$. Its bar complex has a contact-term tower that terminates at degree 4 (shadow class C), so the transferred $A_\infty$-products $m_n$ satisfy $m_n = 0$ for $n \geq 5$. But Koszulness requires $m_n = 0$ for $n \geq 3$.

The vanishing of $m_3$ and $m_4$ is proved by explicit computation: the genus-zero shadow obstructions $o^{(0)}_3 = o^{(0)}_4 = 0$ at generic conformal weight. This establishes $A_\infty$-formality, equivalence (iii). The $\beta\gamma$ system is the simplest family in class C, the class where the tower is finite but nontrivial ($S_3 \neq 0$, $S_4 \neq 0$, $S_r = 0$ for $r \geq 5$).

6.7. **Bershadsky–Polyakov.** The Bershadsky–Polyakov algebra $\mathcal{W}^{(2)}_k(\mathfrak{sl}_3)$ is the non-principal $\mathcal{W}$-algebra obtained by DS reduction of $V_k(\mathfrak{sl}_3)$ with respect to the minimal nilpotent. It has generators at conformal weights $\{1, 3/2, 3/2, 2\}$ and central charge, following Feigin–Kac–Roan–Wakimoto:

$$c(\mathrm{BP}_k) \;=\; 2 - \frac{24(k+1)^2}{k+3}.$$

The formula is singular at the critical level $k = -3 = -h^\vee(\mathfrak{sl}_3)$, where the Sugawara denominator $k + h^\vee$ of the parent $V_k(\mathfrak{sl}_3)$ vanishes; the self-dual point $k = -3$ of the involution $k \mapsto -k - 6$ coincides with this critical locus, and the value $\kappa(\mathrm{BP}_{-3}) = 49/3$ quoted below is the *principal value* of the symmetric limit across the pole, not a regular function value. The polynomial identity $K_{\mathrm{BP}}(k) := c(k) + c(-k-6) \equiv 196$ holds in $\mathbb{Q}(k)$, and $c(k) - 98$ is an odd function of $(k + 3)$.

Koszulness is established through Ext diagonal vanishing (xii): the bigraded Ext groups $\mathrm{Ext}^{p,q}(\omega, \omega)$ are concentrated on the diagonal $p = q$ at generic level. The proof uses the non-principal DS reduction from $V_k(\mathfrak{sl}_3)$: the DS functor sends the Kac–Shapovalov form of the parent to the Bershadsky–Polyakov Shapovalov form, and nonvanishing at generic level gives diagonal concentration.

The modular characteristic is $\kappa(\mathcal{W}_k^{(2)}) = c(k)/6$, with anomaly ratio $\varrho = 1/6$. The Koszul complementarity conductor is $K_{\mathrm{BP}} = \kappa(\mathcal{W}_k^{(2)}) + \kappa(\mathcal{W}_{k'}^{(2)}) = 98/3$, where $k' = -k - 6$ is the dual level. The self-dual level is $k = -3$, at which $\kappa = 49/3$.

## 7. The admissible-level question

The standard landscape verification of §6 treats the universal enveloping VOA $V_k(\mathfrak{g})$ at generic level. At admissible levels $k = -h^\vee + p/q$ (coprime $p \geq h^\vee$, $q \geq 2$), the simple quotient $L_k(\mathfrak{g}) = V_k(\mathfrak{g})/I_k$ is a rational vertex algebra. The question is whether the simple quotient remains Koszul.

### 7.1. **Rank one: $\mathfrak{sl}_2$.**

**Proposition 7.1** ($\mathfrak{sl}_2$ at admissible level)**.** *The simple quotient $L_k(\mathfrak{sl}_2)$ is chirally Koszul at every admissible level $k = -2 + p/q$ with $p \geq 2$, $q \geq 2$,* $\gcd(p, q) = 1$.

*Proof sketch.* The ideal $I_k \subset V_k(\mathfrak{sl}_2)$ is generated by a single null vector at conformal weight $pq$. Since $\dim \mathfrak{sl}_2 = 3$, the PBW spectral sequence for $L_k(\mathfrak{sl}_2)$ has at most two nonzero columns at each conformal weight: the Cartan direction $h$ and the root directions $e, f$. The root generators are killed by $[e, f] = h$ on the $E_1$ page, and the Cartan generator survives as a single class in $H^1$.

The null vector at weight $pq$ introduces a relation in the bar complex, but this relation is confined to a single PBW column: it does not create off-diagonal classes in the $E_2$ page. The PBW spectral sequence collapses at $E_2$ because the dimensional constraint ($\dim \mathfrak{sl}_2 = 3$, hence at most 2 nonzero columns in the relevant range) leaves no room for higher differentials $d_r$ with $r \geq 2$. □

*Remark* 7.2 (Mechanism)*.* The proof for $\mathfrak{sl}_2$ uses the small-rank accident: three generators produce at most two nonzero PBW columns at each weight, and a differential $d_r$ for $r \geq 2$ needs at least $r+1$ nonzero columns as domain and target. At rank one, the room runs out immediately. This is specific to $\mathfrak{sl}_2$; it fails at every higher rank.

### 7.2. **Rank two and beyond: the obstruction.**

**Conjecture 7.3** (Admissible Koszul rank obstruction)**.** *For every simple Lie algebra $\mathfrak{g}$ of rank $r \geq 2$ and every admissible level $k = p/q - h^\vee$ with denominator $q \geq 3$, the simple quotient $L_k(\mathfrak{g})$ has $\dim H^2(\bar{B}^{\mathrm{ch}}(L_k(\mathfrak{g}))) = \mathrm{rank}(\mathfrak{g})$. In particular, $L_k(\mathfrak{g})$ is not chirally Koszul.*

The mechanism is the abelian Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$. On the $E_1$ page of the PBW spectral sequence for $L_k(\mathfrak{g})$, the Poisson differential kills the root generators via $[e_\alpha, f_\alpha] = h_\alpha$, but the Cartan generators $h_1, \ldots, h_r$ survive because $\mathfrak{h}$ is abelian. When $r = 1$, the surviving class is confined to a single column and cannot support a higher differential. When $r \geq 2$, the surviving $r$-dimensional space admits nontrivial higher differentials sourced by the multi-weight coupling between null vectors at different roots: the null vectors interact through the Cartan matrix, and the interactions generate off-diagonal bar cohomology.

*Remark* 7.4 (Evidence)*.* For $\mathfrak{g} = \mathfrak{sl}_3$ at admissible level with $q = 3$: $\dim H^2(\bar{B}^{\mathrm{ch}}(L_k(\mathfrak{sl}_3))) = 2$, confirmed by direct computation. The two classes correspond to the two simple roots. For $\mathfrak{g} = \mathfrak{sl}_4$: predicted $H^2 = 3$. The pattern is $\dim H^2 = \mathrm{rank}(\mathfrak{g})$.

*Remark* 7.5 (Physical content)*.* Admissible-level simple quotients are the vertex algebras of rational conformal field theories (Arakawa). If Conjecture 7.3 holds, then bar–cobar inversion fails for these quotients at rank $\geq 2$: the null vectors that enforce rationality of the representation category simultaneously obstruct PBW concentration. Koszulness and rationality would stand in tension for rank $\geq 2$: the algebraic optimality of the bar complex (Koszulness) is incompatible with the representation-theoretic rigidity of the simple quotient (rationality). At rank one, both coexist.

## 8. The shadow algebra and BV/BRST

The Koszul property controls the bar complex at genus zero. At higher genus, the bar differential squares to a nonzero curvature $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g$, and the Maurer–Cartan element $\Theta_{\mathcal{A}}$ is the datum that measures the obstruction to extending genus-zero results across the modular tower. This section records two structures that accompany the Koszul property: the shadow algebra (the graded Lie algebra controlling the MC recursion) and the BV/bar comparison (the genus-by-genus identification with the BRST complex).

### 8.1. **The shadow algebra.**

The shadow invariants $S_r(\mathcal{A})$ for $r \geq 2$ are the degree-$r$ projections of $\Theta_{\mathcal{A}}$. They do not form a ring: the natural algebraic structure is a *graded Lie bracket*, inherited from the convolution Lie algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$.

**Definition 8.1** (Shadow algebra)**.** The *shadow algebra* of a chiral algebra $\mathcal{A}$ is the bigraded Lie algebra

$$\mathcal{A}^{\mathrm{sh}} = \bigoplus_{r\geq 2,\ g\geq 0} \mathcal{A}^{\mathrm{sh}}_{r,g}$$

where $\mathcal{A}^{\mathrm{sh}}_{r,g}$ is the degree-$r$, genus-$g$ component of the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$. The Lie bracket $[-,-]^{\mathrm{sh}}\colon \mathcal{A}^{\mathrm{sh}}_{r_1,g_1} \otimes \mathcal{A}^{\mathrm{sh}}_{r_2,g_2} \to \mathcal{A}^{\mathrm{sh}}_{r_1+r_2-2,g_1+g_2}$ descends from the convolution bracket and drives the Maurer–Cartan recursion: $S_{r+1}$ is determined by $S_2, \dots, S_r$ through

$$d\, S_{r+1} + \tfrac{1}{2} \sum_{\substack{r_1+r_2=r+2 \\ r_1,r_2\geq 2}} [S_{r_1}, S_{r_2}]^{\mathrm{sh}} = 0.$$

*Warning* 8.2 (Not a ring)*.* The shadow algebra is a graded Lie algebra, *not* a graded-commutative ring. There is no shadow product $S_r \cdot S_s$; there is a shadow bracket $[S_r, S_s]^{\mathrm{sh}}$. The distinction is structural: the Jacobi identity of $[-,-]^{\mathrm{sh}}$ controls the solvability of the MC recursion, and the depth classification G/L/C/M records where this recursion first produces nonzero data.

The G/L/C/M classification follows from the shadow algebra:

- Class G ($r_{\max} = 2$): $S_r = 0$ for $r \geq 3$. The shadow algebra is abelian. Heisenberg, lattice algebras.
- Class L ($r_{\max} = 3$): $S_3 \neq 0$, $S_r = 0$ for $r \geq 4$. The cubic shadow detects the Lie bracket. Affine Kac–Moody at non-critical level.
- Class C ($r_{\max} = 4$): $S_3 \neq 0$, $S_4 \neq 0$, $S_r = 0$ for $r \geq 5$. Contact-term termination. The $\beta\gamma$ system.
- Class M ($r_{\max} = \infty$): $S_r \neq 0$ for infinitely many $r$. Virasoro, $\mathcal{W}$-algebras.

Every class is chirally Koszul. The depth classification is orthogonal to Koszulness: it measures complexity within the Koszul world, not failure of Koszulness.

8.2. **BV/bar identification in the coderived category.** The BV/BRST complex $C^{\bullet}_{\mathrm{BV}}(\mathcal{A}, \Sigma_g)$ on a genus-$g$ surface $\Sigma_g$ computes the same factorization homology as the bar complex $B^{(g)}(\mathcal{A})$, but through a different chain model: the BV propagator is the Green's function on $\Sigma_g$, while the bar propagator is the logarithmic form $d \log E(z,w)$ (weight 1, the prime form logarithmic derivative).

**Theorem 8.3** (BV=bar in the coderived category)**.** *Let $\mathcal{A}$ be a chirally Koszul algebra.*

(i) *For every genus $g \geq 0$, $C^{\bullet}_{\mathrm{BV}}(\mathcal{A}, \Sigma_g) \simeq_{D^{\mathrm{co}}} B^{(g)}(\mathcal{A})$ in the coderived category of curved factorization models.*

(ii) *If $\mathcal{A}$ is class G or class L, the comparison morphism $f_g$ is a chain-level weak equivalence.*

(iii) *If $\mathcal{A}$ is class C and harmonic decoupling holds, $f_g$ is a chain-level weak equivalence.*

(iv) *If $\mathcal{A}$ is class M, the comparison cone is coacyclic: $f_g$ is a coderived equivalence but* not *a chain-level quasi-isomorphism. The harmonic discrepancy at degree $r$ is $\delta_r^{\mathrm{harm}} = c_r(\mathcal{A}) \cdot m_0^{\lfloor r/2 \rfloor - 1}$, a sum of curvature powers whose cone is exact in the coderived sense.*

*Proof sketch.* The comparison morphism $f_g$ is constructed by replacing the BV propagator with its Hodge decomposition relative to the bar propagator. At genus 0, the Hodge decomposition is trivial (no harmonic forms on $\mathbb{P}^1$), and $f_0$ is a chain-level quasi-isomorphism for all classes.

At genus $g \geq 1$, the Hodge decomposition produces a harmonic correction proportional to the curvature $m_0 = \kappa(\mathcal{A}) \cdot \omega_g$. For class G, $\kappa = k$ and $S_r = 0$ for $r \geq 3$: the harmonic correction at each degree involves only the curvature itself, and the comparison is exact. For class L, the cubic shadow contributes a single additional correction at degree 3, which is killed by the Jacobi identity of the shadow bracket.

For class M, the harmonic corrections are nonzero at every degree $r \geq 4$. Each correction $\delta_r^{\mathrm{harm}}$ is a polynomial in the curvature $m_0$; the sum defines the comparison cone. This cone is *not* acyclic in the ordinary sense (the harmonic terms persist at each degree), but it *is* coacyclic: it belongs to the acyclic class of the coderived category. The proof uses the characterization of coacyclic objects as those whose totalizations are contractible in the complete filtered sense [24, §3.7]. □

*Remark* 8.4 (Class M: chain-level false). The failure of the chain-level BV/bar comparison for class M is genuine: it is not an artifact of a suboptimal chain model. The Virasoro algebra has $\delta_4^{\mathrm{harm}} = c_4(\mathrm{Vir}_c) \cdot m_0 \neq 0$ for $c \neq 0$, and $c_4$ is determined by the shadow invariant $S_4 = 10/[c(5c+22)]$. Since $S_4 \neq 0$ for generic $c$, the harmonic discrepancy is nonzero, and no chain-level comparison exists. The coderived category is the correct ambient category for BV/bar at class M.

*Remark* 8.5 (Scope summary). BV=bar in the coderived category is proved unconditionally for all chirally Koszul algebras. The chain-level strengthening holds for class G and class L (unconditionally), for class C (conditionally on harmonic decoupling), and fails for class M. The coderived formulation is the natural one: it accommodates the curvature $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ without requiring chain-level cancellation.

## 9. First-principles triples

Each characterization that was historically misidentified contains a *ghost of a correct theorem*: what the wrong statement got right, what it got wrong, and the precise correct statement. We record the three triples that organized this paper.

Triple 1: meta-theorem vs moduli scheme. *Right:* the classical list of equivalent conditions. The ten classical characterizations truly do detect the same property (chiral Koszulness) when checked at the KZ associator. *Wrong:* reading the list as a *definitional* redundancy ("equivalence as tautology"). This misses the Grothendieck–Teichmüller structure: each condition is adapted to a specific associator, and $\mathrm{GRT}_1$ acts on the set of conditions by permutation. *Correct:* the conditions are charts on a $\mathrm{GRT}_1$-torsor $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$ (Definition 2.1, Theorem 2.2). The equivalences are transition maps; the redundancy is organized by the torsor structure; and the full atlas has fourteen charts, not ten.

Triple 2: Virasoro circularity vs Hodge–Betti chart. *Right:* Virasoro at generic $c$ is chirally Koszul. *Wrong:* the classical proof uses bar–cobar inversion to prove a theorem whose conclusion is bar–cobar inversion. This is circular. *Correct:* the Hodge–Betti chart $U_{12}$ detects Koszulness via Feigin–Fuchs BGG resolution and Kac determinant non-degeneracy, without invoking bar–cobar (Theorem 2.9). The bar complex is a consumer of the BGG–Kac data; the arrow goes Feigin–Fuchs $\to$ bar, never bar $\to$ bar.

Triple 3: Yangian associative-to-chiral lift. *Right:* $Y_\hbar(\mathfrak{g})$ is associatively Koszul (Polishchuk–Positselski via RTT PBW). *Wrong:* reading the classical proposition as asserting chiral Koszulness. The classical proof is strictly associative; its chiral content is stated but not produced. *Correct:* there is an open embedding $\mathcal{M}^{\mathrm{assoc}}_{\mathrm{Kosz}}(Y_\hbar) \hookrightarrow \mathcal{M}^{\mathrm{chiral}}_{\mathrm{Kosz}}(Y_\hbar)$ (Theorem 2.11). The honest open question is chart-surjectivity: whether every chiral chart lifts an associative one. At $\mathrm{sl}_2$ it is an isomorphism; at higher rank, it is genuinely open.

The Koszul locus is the locus on which the bar construction is invertible, the spectral sequence collapses, the resolution is linear, and a single scalar controls the Maurer–Cartan element at degree two. Fourteen charts on the $\mathrm{GRT}_1$-torsor $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$ detect this locus; their equivalence is the torsor structure, and their content is the content of this paper. Every standard chiral algebra lies on the Koszul locus. Shadow depth (the G/L/C/M classification) is an orthogonal invariant: it classifies complexity within the Koszul world, not proximity to the Koszul boundary. The admissible-level question (whether simple quotients at rank $\geq 2$ remain Koszul) and the $\mathcal{D}$-module purity converse (whether Koszulness implies Hodge-theoretic purity beyond affine Kac–Moody) are the two open frontiers where the boundary of the Koszul locus has not yet been mapped.

## References


[1] A. Alekseev and C. Torossian. The Kashiwara–Vergne conjecture and Drinfeld's associators. *Ann. of Math.* **175** (2012), 415–463.

[2] A. Beilinson and V. Drinfeld. *Chiral Algebras.* AMS Colloquium Publications, vol. 51, 2004.

[3] P. Deligne. Le groupe fondamental de la droite projective moins trois points. In *Galois Groups over* , MSRI Publ. 16, Springer, 1989, pp. 79–297.

[4] V. G. Drinfeld. On quasitriangular quasi-Hopf algebras and a group closely connected with Gal(/). *Leningrad Math. J.* **2** (1991), 829–860.
[5] B. Enriquez. Elliptic associators. *Selecta Math.* **20** (2014), 491–584.
[6] B. L. Feigin and D. B. Fuchs. Representations of the Virasoro algebra. In *Representation of Lie Groups and Related Topics*, Adv. Stud. Contemp. Math. **7**, Gordon and Breach, 1990, pp. 465–554.
[7] H. Furusho. Pentagon and hexagon equations. *Ann. of Math.* **171** (2010), 545–556.
[8] V. G. Kac. *Infinite Dimensional Lie Algebras*. Third edition, Cambridge University Press, 1990.
[9] M. Kontsevich. Vassiliev's knot invariants. In *I. M. Gel'fand Seminar*, Adv. Soviet Math. **16**, AMS, 1993, pp. 137–150.
[10] M. Kontsevich. Operads and motives in deformation quantization. *Lett. Math. Phys.* **48** (1999), 35–72.
[11] A. Rocha-Caridi and N. R. Wallach. Characters of irreducible representations of the Virasoro algebra. *Math. Z.* **185** (1984), 1–21.
[12] D. Tamarkin. Formality of chain operad of little discs. *Lett. Math. Phys.* **66** (2003), 65–72.
[13] A. Beilinson, V. Ginzburg, and W. Soergel. Koszul duality patterns in representation theory. *J. Amer. Math. Soc.* **9** (1996), 473–527.
[14] K. Costello and O. Gwilliam. *Factorization algebras in quantum field theory*, vols. I–II. Cambridge University Press, 2017/2021.
[15] J. Francis and D. Gaitsgory. Chiral Koszul duality. *Selecta Math. (N.S.)* **18** (2012), 27–87.
[16] W. Fulton and R. MacPherson. A compactification of configuration spaces. *Ann. of Math.* **139** (1994), 183–225.
[17] H. Garland and J. Lepowsky. Lie algebra homology and the Macdonald–Kac formulas. *Invent. Math.* **34** (1976), 37–76.
[18] D. Gaitsgory and N. Rozenblyum. *A study in derived algebraic geometry*. American Mathematical Society, 2017.
[19] J. Lurie. *Higher Algebra*. 2017. Available at https://www.math.ias.edu/~lurie/.
[20] V. Kac and D. Kazhdan. Structure of representations with highest weight of infinite-dimensional Lie algebras. *Adv. in Math.* **34** (1979), 97–108.
[21] B. Keller. $A$-infinity algebras, modules and functor categories. In *Trends in Representation Theory of Algebras and Related Topics*, Contemp. Math. **406** (2006), 67–93.
[22] J.-L. Loday and B. Vallette. *Algebraic Operads*. Grundlehren der math. Wissenschaften, vol. 346, Springer, 2012.
[23] R. Lorgat. *Modular Koszul Duality*. In preparation, 2026.
[24] A. Polishchuk and L. Positselski. *Quadratic Algebras*. University Lecture Series, vol. 37, AMS, 2005.
[25] S. Priddy. Koszul resolutions. *Trans. Amer. Math. Soc.* **152** (1970), 39–60.
[26] M. Saito. Modules de Hodge polarisables. *Publ. Res. Inst. Math. Sci.* **24** (1988), 849–995.
[27] M. Semenov-Tian-Shansky. What is a classical $r$-matrix? *Funct. Anal. Appl.* **17** (1983), 259–272.

Perimeter Institute for Theoretical Physics, Waterloo, ON N2L 2Y5, Canada
*Email address*: rlorgat@perimeterinstitute.ca